

ATLAS: Agenda Tracking and Linking through Adaptive Selection for Streaming Clinical Dialogues

Anonymous ACL submission

Abstract

Clinical conversations contain multiple concerns and questions that emerge, recur, and may remain unresolved. We introduce **ATLAS** (Agenda Tracking and Linking through Adaptive Selection), a streaming framework that updates a structured agenda state from each utterance under bounded context. ATLAS extracts typed events, associates them with dynamic agendas, links open questions to exact answer utterances or abstains, and derives whether each agenda is unaddressed, addressed, or fully addressed. A confidence router accepts high-confidence NLI candidates, rejects low-confidence ones, and sends only uncertain cases to an LLM verifier. **On a held-out 140-conversation split, confidence routing reaches 0.775 patient and 0.812 physician exact-link accuracy while reducing verifier calls by about 83% relative to always verifying.** Controlled synthetic conversations augment sparse agenda and unanswered-question patterns. Bounded, learned context selection matches or beats retaining far more context, so task-relevant selection matters more than context quantity.

1 Introduction

Conversational AI is moving from query-driven interaction toward ambient systems that operate in the settings where conversations happen (Reddi, 2025). Clinical medicine is a leading example. Ambient documentation systems already listen to the visit and generate clinical notes at deployment scale (Tierney et al., 2024; Kanaparthi et al., 2025), yet their primary output is a clinical note rather than a live structured account of the encounter.

The clinical-communication literature identifies what such a system should track: *agenda setting* (eliciting, organizing, and revisiting the patient’s concerns) (Sierpe et al., 2024). In practice, physicians redirect patient openings after roughly 23 seconds (Marvel et al., 1999), and one observational study found agenda elicitation in only 36%

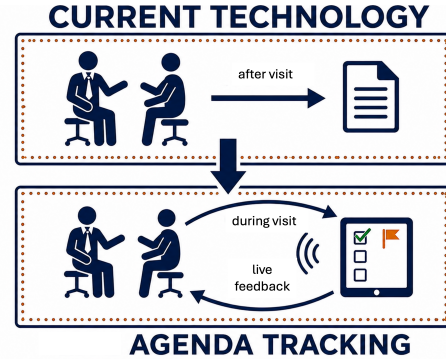
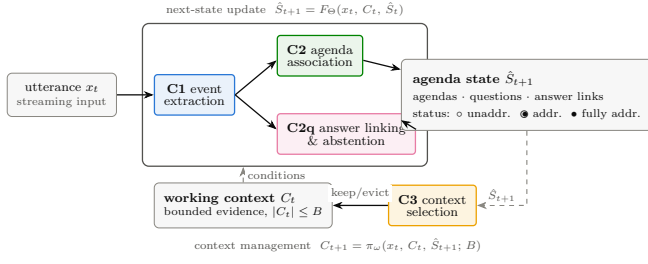


Figure 1: **Real-time agenda tracking.** Current ambient documentation technology returns feedback only as a note after the visit. Live agenda tracking surfaces open concerns during the visit, supporting the patient-provider interaction as it happens.

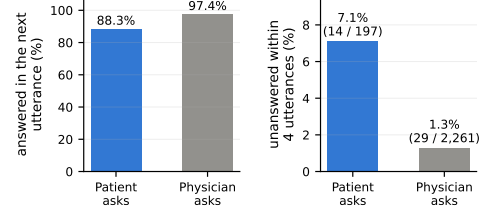
of encounters (Singh Ospina et al., 2019). A system that tracks the agenda in real time (which concerns are open, which have been discussed, which questions have received answers) would support agenda setting during the visit itself (Figure 1).

The encounter itself is structurally imbalanced: physicians open 89.3% of agendas, and 98.7% of their questions are answered within four utterances, while patient questions wait longer and are far more likely to ~~never be answered~~ **go unanswered within that window** (Figure 2b). Patient questions are rare, and ~~when one goes unanswered it usually fails silently~~ **an unanswered one can fail silently**, deflected by a counter-question, a backchannel, or a deferral the visit never returns to, so neither party notices. A live system can fill this gap by tracking which concerns are open and whether each question was actually answered, not merely responded to.

The task is therefore not encounter summarization but *live agenda-state tracking*: at every utterance, the system must (i) identify agenda-relevant events; (ii) decide whether each opens a new agenda or belongs to an existing one; (iii) link each



(a) ATLAS overview. Each utterance x_t updates the agenda state $\hat{S}_t$, and a learned policy maintains the bounded working context C_t (§3).



(b) Response-time asymmetry (frozen lookahead-4 human annotations): physician questions are answered at the next utterance 97.4% of the time, while patient questions are 5.5 $\times$ as likely to go unanswered (7.1% vs. 1.3%).

Figure 2: ATLAS system overview (a) and the motivating patient–physician response asymmetry (b).

answered question to the *exact* answering utterance; (iv) abstain when no candidate answers it; (v) update every agenda’s status; and (vi) retain only the context needed later.

These requirements are hard for conventional pipelines. Topical similarity is not answerhood, since a semantically close reply may defer the question, acknowledge it, or answer a different one. Agenda identities are open-world, so no fixed slot schema applies, and an answer may arrive several turns after its question. Retaining all context does not help, because models attend *unreliably* to long contexts (Liu et al., 2024), and in our experiments retaining more context often hurts. Errors also propagate. An agenda opening missed at extraction blocks every later association and linking decision.

We introduce **ATLAS** (Agenda Tracking and Linking through Adaptive Selection), a streaming framework that treats clinical agenda understanding as structured *next-state prediction*: at each turn, the agenda state is updated from the current utterance, the previous state, and a bounded working context (Figure 2a). Unlike slot-based dialogue-state tracking, the state is hierarchical and open-world, every intermediate state is scored as the visit unfolds, and the update is coupled to a learned bounded context (§3). An extraction stage identifies typed events, an association stage attaches them to dynamically created agendas (opening, re-visiting, and merging them as the conversation develops), and a linking stage connects open questions to exact answer utterances or abstains, with an NLI model settling clear cases and an LLM verifier consulted only in an uncertain confidence band. A learned selection policy decides which condensed evidence stays in the working context. Each agenda’s status (unaddressed, addressed, or fully addressed) is derived deterministically from

recorded events and links, so every reported state is traceable.

Our experiments show that selectivity, not scale, makes online agenda tracking reliable. Confidence routing retains competitive exact-link accuracy at a fraction of verifier-only cost (§7.3), bounded learned context selection matches or beats retaining far more context (§7.4), and the remaining end-to-end limit is the agenda openings the extractor misses (§7.2, §7.5).

Contributions.

- Formulation:** online agenda tracking as structured next-state prediction over an open-world hierarchical state, scored at every intermediate turn, with the update coupled to a learned bounded context.
- System:** a streaming pipeline with typed event extraction, dynamic agenda association, exact answer linking whose calibrated abstention sends only an uncertain band to an LLM verifier, and attention-supervised context selection.
- Corpus:** 207 clinical conversations with corrected speaker attributions and reference labels for events, parent links, and exact answers or abstentions, plus separately evaluated synthetic augmentation.

2 Related Work

Clinical agenda setting and ambient clinical intelligence. Agenda setting structures patients’ concerns (Sierpe et al., 2024), but patients’ agendas are elicited inconsistently in practice (Marvel et al., 1999; Singh Ospina et al., 2019). Ambient documentation has reached deployment scale (Tierney et al., 2024; Kanaparthi et al., 2025), supported by dialogue corpora and synthetic generation (Yim

et al., 2023; Wang et al., 2024), but its output is the clinical note; ATLAS instead maintains an online structured account during the encounter.

Dialogue-state tracking and answer linking.

Task-oriented dialogue-state tracking maintains user goals over a predefined slot schema, recently zero-shot with LLMs (Li et al., 2024b; Feng et al., 2023). Agenda identities are instead *open-world*, and status is derived from the tracked state rather than stored in a slot (§3). Answerability research established no-answer detection over fixed reference texts and conversational dialogue (Rajpurkar et al., 2018; Choi et al., 2018), with NLI models as answer verifiers (Williams et al., 2018; Chen et al., 2021a), but makes a one-shot decision. Our linker operates online over standing questions, emits the exact answer-utterance identifier or abstains, and routes only an uncertain confidence band to an LLM verifier.

Context selection and memory management.

KV-cache compression, prompt compression, and agent-memory systems manage long histories at the model and system levels (Xiao et al., 2024; Jiang et al., 2023; Packer et al., 2023). Our context selector operates at the *application* level, and our results (§7.4) show that a policy can retain context yielding fluent summaries while evicting the rare evidence future transitions depend on.

An extended survey of each area appears in Appendix B.

3 Problem Formulation and Challenges

Agenda tracking departs from task-oriented dialogue-state tracking (§2) in three ways. The state is *open-world and hierarchical*: no slot schema exists, and every child attaches to a parent the tracker itself predicted. Prediction is *streaming*: every intermediate state is scored as the encounter unfolds, so a correct final state does not excuse a late opening or a late answer link. Finally, the transition is *context-coupled*: what the tracker can infer at turn t depends on what a learned bounded policy chose to remember, so state tracking and context management cannot be optimized or evaluated in isolation.

Streaming next-state prediction. A clinical encounter arrives as a stream $D = (x_1, \dots, x_T)$, where each x_t carries the utterance text, its speaker, and its turn index. ATLAS maintains two coupled objects across the stream. The first is a persistent

structured agenda state, a two-level hierarchy over everything the conversation has raised,

$$\hat{S}_t = (\mathcal{A}_t, \{(E_t(a), Q_t(a), R_t(a))\}_{a \in \mathcal{A}_t}, G_t), \quad (1)$$

where $\mathcal{A}_t$ is the open-world set of agendas opened by turn t , each agenda a carries its typed events $E_t(a)$, its questions $Q_t(a)$, and its exact question-to-answer links $R_t(a) \subseteq Q_t(a) \times \{1, \dots, t\}$, and G_t holds global events attached to no agenda. The upper level is the agendas themselves, each opened by an `agenda_item` event. The lower level is their children: every remaining event type (defined below) attaches to a parent agenda, except `social_history`, which is global. A question is never itself an agenda item, so linking its answer does not by itself resolve the parent concern. The second is a bounded working context C_t : the condensed textual evidence that the next model call is allowed to see. Each extracted event carries a one-sentence summary emitted at extraction time. C_t pools these summaries across all event types, question and global events included, and the typed split of the state below does not partition the buffer. At every turn the system first updates the state,

$$\hat{S}_{t+1} = F_\Theta(x_t, C_t, \hat{S}_t), \quad (2)$$

and the context manager then decides what evidence the following update will condition on,

$$C_{t+1} = \pi_\omega(x_t, C_t, \hat{S}_{t+1}; B), \quad |C_{t+1}| \leq B. \quad (3)$$

The two objects fail differently: errors in the agenda state are wrong or missing links, while errors in the working context are evictions of evidence a future transition depends on.

A fixed deterministic operator ψ constructs states from annotated events together with their annotated parent links and exact answer links. Replaying these annotations through it induces the reference trajectory $S_t^* = \psi(E_{\leq t}^*)$, where $E_{\leq t}^*$ denotes the annotation record up to turn t , so reference states are never separately annotated, and comparing state trajectories isolates the errors of the learned components. Each predicted transition is scored against S_{t+1}^* as the stream advances. An error at turn t , such as a missed agenda opening or a wrong parent, therefore propagates into all later states.

Typed event extraction. Each utterance is converted into zero or more typed

events, $\hat{E}_t = f_\theta(x_t, C_t, \hat{S}_t)$, over six types: agenda_item, question, detail, medication, social_history, and follow_up. An event records its type, its source utterance index, and a one-sentence summary, so it is at once a structured entry for the state and the condensed text that enters the working context. An utterance with no clinical content produces an empty event set. Extraction is multi-label: a single annotated utterance can carry several distinct events, so the extractor cannot commit to one event per turn.

Agenda association and state construction.

Each extracted event must be attached to an agenda, $\alpha_t(e) \in \mathcal{A}_t \cup \{\text{NEW, GLOBAL}\}$. Since agenda identities are open-world, association subsumes four decisions: whether the event introduces a new agenda, which existing agenda is its parent, whether a dormant agenda is being revisited, and whether two apparently distinct agendas should be merged.

Exact answer-utterance linking and abstention.

For each open question q , the candidates $\mathcal{U}_t(q)$ are the opposite-speaker utterances within a bounded lookahead K , set empirically at the elbow of the annotated answer-position distribution (§7.3). **The bound plays the role for linking that the budget B plays for context: it turns an unbounded streaming decision into a finite one. Without it, unansweredness is undecidable while the encounter is running, since a later answer can always arrive, and the candidate set grows with the transcript. With it, a question with no linked answer after K opposite-speaker utterances resolves to unanswered during the visit, when circling back is still possible, and every linking decision costs a fixed number of comparisons. The price is that answers arriving beyond the window are unlinkable by construction (§7.3).** The linker returns an exact answer utterance $\hat{u}(q) \in \mathcal{U}_t(q)$ or abstains ($\hat{u}(q) = \emptyset$), and a question counts as answered exactly when such a link exists. A returned link is what populates the state’s link component: it is recorded as $(q, j) \in R_{t+1}(a)$, where $x_j = \hat{u}(q)$ and a is the question’s parent agenda.

Abstention is a required output. Some patient questions [delete: never receive an answer]**receive no answer within the window, and a smaller set receives none anywhere in the encounter (§7.3),** and unanswered physician questions typically draw topically adjacent but off-target replies (§9). The linker must therefore be able to predict that no

candidate answers q .

Agenda-status semantics. Status is derived from the state rather than predicted. An agenda is *addressed* once any evidence of engagement exists and *fully addressed* when every associated question has an answer link, giving $z_t(a) \in \{\text{UNADDRESSED, ADDRESSED, FULLYADDRESSED}\}$ as a deterministic function of recorded events and links (formal definition in Appendix H).

4 The ATLAS Framework

ATLAS realizes the transition of §3 with four components (Figure 2a). C1 extracts typed events from each utterance; C2 associates them with dynamically created agendas and maintains the persistent agenda state; C2q resolves each open question to an exact answer utterance or an abstention; together they implement the state update F_Θ . C3 implements the context policy π_ω , deciding which condensed evidence remains in the bounded working context that conditions the next update. [Internals of the confidence router and the selection policy are shown in Figure 3.](#)

4.1 C1: Typed event extraction

C1 (Agenda_LLM) converts each utterance into a set of typed events in a single pass over the stream. Its input is a pipe-delimited buffer of its own previous summaries ([Context]) and the current speaker-prefixed [Utterance], never the raw transcript of earlier turns, which keeps the [delete: on-device] context budget small. This instantiates $f_\theta(x_t, C_t, \hat{S}_t)$ of §3, with the state entering through the retained summaries.

The output is multi-label (§3): an utterance may carry several typed events, each with a one-sentence summary and the identifier of its source utterance so downstream components can recover the turn. One further label, irrelevant, marks utterances with no clinical content.

4.2 C2: Agenda construction and association

C2 (State_Tracker) organizes the C1 event stream into agenda nodes. Each node represents one clinical topic. For an incoming topic, C2 first checks existing nodes for a case-insensitive name match. If the names do not match, it uses embedding similarity and merges the topic with a node when their cosine similarity is above 0.85. If neither match applies, C2 creates a new node.

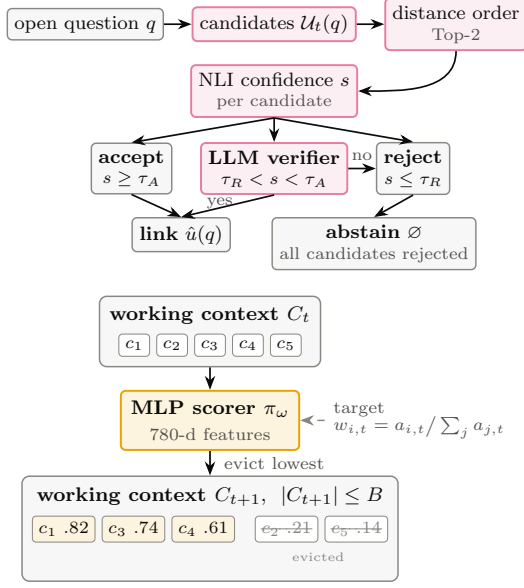


Figure 3: Component internals. **Top (C2q)**: candidates within lookahead $K=4$, kept in distance order (Top-2), are routed by NLI confidence to accept ($s \geq \tau_A$), reject ($s \leq \tau_R$), or LLM verification in between. **Bottom (C3)**: an MLP scores buffered summaries and evicts the lowest until $|C_{t+1}| \leq B$.

C2 links each question, detail, medication, or follow-up event to a parent topic in three steps. It first checks for a **same-utterance match**. If that fails, it applies a **semantic match** when similarity is above 0.25. If both fail, it uses an **active-agenda fallback** to the most recently active topic. This fallback assumes speakers usually stay on topic. **When no agenda node has been opened yet, the fallback has no target and the event is left unlinked.**

Nodes start as *mentioned*. A node becomes *discussed* once both speakers have contributed. Nodes with follow-up events are also marked *resolved*. For reporting, these nodes also count as *discussed*. **These internal states are runtime diagnostics scored by the resolution metric of §7.2. The reported agenda statuses (Unaddressed, Addressed, FullyAddressed) are derived separately from recorded details and answer links (§3, Appendix H) and do not depend on them.** Social-history events are stored separately and are not linked to a topic. C2 assigns events to topics. C2q determines whether questions have been answered.

4.3 C2q: Candidate retrieval and edge scoring

Candidates: When C1 emits a question, the system registers it as open. Its candidates are the bounded set of §3 **with lookahead $K = 4$** . The linker selects one as the answer or abstains, never

generating or paraphrasing an answer. **Links are recorded at the utterance level, and short replies (“yeah”, a bare number) stay eligible without an extracted event, so a direct answer is never lost to a failed extraction.** Candidates are ordered by dialogue distance with the nearest two retained (Top-2), an ordering chosen over the HGB edge score for balanced accuracy (§8).

Confidence router: **The router treats answerhood as an entailment decision rather than a similarity threshold.** Scoring every question–candidate pair with a large model is too expensive for online use. The router therefore uses an NLI cross-encoder as a cheaper first step. The cross-encoder assigns a confidence score s to each retained candidate. A candidate with $s \geq \tau_A = 0.75$ is accepted directly. A candidate with $s \leq \tau_R = 0.10$ is rejected. Only candidates in the uncertain band $\tau_R < s < \tau_A$ are passed to the C1 (Agenda_LLM) verifier. A question is reported as unanswered if all its candidates are rejected. The NLI scorer is cross-fitted over five encounter-family folds (seed 42). The folds have no family or conversation overlap. Every score used at evaluation time is therefore fold-external.

4.4 C3: Attention-supervised MLP policy

C3 (Context_Manager) manages the buffer of prior C1 summaries on which C1 conditions, under a fixed token budget. When the buffer exceeds the budget, C3 scores each buffered source summary and evicts the lowest-scoring ones **until the buffer fits within the budget**. It selects among existing summaries and never generates or rewrites them. **The budget is nominal at the summary boundary: a single retained summary longer than B tokens is kept whole rather than truncated, so the buffer can briefly exceed B (observed overruns at $B = 64$ reach at most 110 tokens, Appendix R).** We evaluate budgets $B \in \{64, 100, 128, 256\}$.

Pairwise features and MLP scorer. Each source–target pair is represented by a 780-dimensional vector consisting of two frozen MiniLM (Wang et al., 2020) 384-dimensional summary embeddings and 12 auxiliary features covering position, recency, speaker, length, medical-entity count, and event type. **A three-layer MLP with dimensions (780-128-64-1) assigns a scalar score to each source–target pair, using ReLU activations between the hidden layers (details in Appendix S).** C3 scores only prior buffered summaries against the current-step output; **the current output**

serves only as the target and is not considered for eviction.

Attention-derived training targets. For each source summary $c_i \in C_t$, the training target is derived from the attention that C1 assigns to it while generating the current target summary y_t . The attention contribution $a_{i,t}$ is computed by averaging attention over the generated output positions and attention heads, summing it over the tokens of c_i , and then averaging across model layers. It is normalized across the buffered source summaries as

$$w_{i,t} = \frac{a_{i,t}}{\sum_{j \in C_t} a_{j,t}},$$

and converted into a within-timestep percentile rank. This rank-based target represents the relative utilization of each source summary and is used to train the MLP scorer.

5 Dataset and Annotation

5.1 Dialogue corpus

The corpus comprises 207 standardized clinical conversations derived from ACI-BENCH (Yim et al., 2023). Relative to the source transcripts, it corrects speaker-attribution errors (which corrupt opposite-speaker candidate retrieval) and adds an annotation layer designed for the agenda-setting task.

Across the corpus’s 11,478 utterances, agenda openings are sparse (403 agenda_item labels, about two per conversation), so the tracker gets few chances to open each agenda and a missed opening is costly. The question set is heavily skewed by speaker role (2,261 physician versus 197 patient questions), which is why the two roles are evaluated separately. Patient questions carry human reference labels, and the physician-question reference set is complete for answerability, with the unanswered subset preliminary. Splits and further statistics appear in §6 and Appendix F.

Annotation pipeline. Annotations are produced in a two-stage teacher-validate loop driven by Gemini 3.1 Flash Lite, with the reference clinical note as the primary annotation criterion. A [delete: 5% second-pass validation accepts 88.6% of sampled annotations]10% second-pass validation accepts 92.1% of 1,197 sampled utterances, rejected annotations were corrected manually, and three human annotators performed a final validation pass over the corpus. A two-annotator agreement pilot and a

corpus-wide teacher-versus-reference comparison appear in Appendix E. Schema, guidelines, and statistics are in Appendices E and F.

5.2 Annotation layers and synthetic data

Each conversation carries six aligned annotation layers: utterance-level event types, normalized one-sentence event summaries, agenda identifiers with parent links, open questions, exact answer-utterance identifiers or explicit no-answer labels, and derived agenda statuses. The layers mirror the pipeline, so the same annotation supports both component-wise evaluation and the end-to-end error analysis. The status layer is derived deterministically from details and answer links rather than labeled independently (§3).

Controlled synthetic generation complements the human annotations with conversations covering rare patterns (e.g., implicit and multi-utterance answers, topically similar non-answers, unanswered questions). Synthetic conversations serve augmentation, stress testing, and ablation only. They never enter the human-annotated denominator and are not evidence of cross-domain generalization. The human corpus contains zero utterance-level question_unanswered [delete: instances]labels (the unanswered questions of §7.3 are derived from the separate answer-link layer), so a constrained pipeline generates three implicit non-answer mechanisms across the 100 synthetic conversations, with automated validation (Appendix F).

6 Experimental Setup

Data splits and evaluation protocol. All separation is at conversation level. After alias normalization, the 207 challenge conversations correspond to 207 distinct encounter families; a post-hoc audit found no family crossings in the existing 67/20/120 train/validation/test split. C1 extraction is evaluated over the pooled validation and test utterances ($n = 7,735$). The C1-v3 verifier was trained on the 67 training conversations, so all reported C2q test results use the remaining 140 conversations. The 20 validation conversations used for C1 model selection are included both in the pooled C1 evaluation and in the 140-conversation C2q split, so these evaluations are held out from verifier training but not from C1 model selection. C2q follows an evaluation protocol frozen before scoring. Its NLI scores are fold-external, from the encounter-family cross-fitting of §4.3. Reported C2q metrics use

human-annotated question roots and distance-order Top-2 candidates, excluding upstream C1 predictions. The routing thresholds are fixed in advance and never tuned on scored test rows.

C1 training configuration. C1 fine-tunes Llama-3.2-3B (Meta AI, 2024) with 4-bit QLoRA (Dettmers et al., 2023), with the configuration (learning rate 2×10^{-4} , rank 16) selected by validation macro-F1 from a sweep over three learning rates and two adapter ranks (Figure 11, appendix). Training data combines the 67 real training conversations with 100 synthetic conversations, expanded to 16,436 examples by multi-label oversampling, context-erased copies, and irrelevant-ratio supplementation (full recipe in Appendix S).

Evaluation modes. Each results subsection isolates one source of error. C1 extraction is scored intrinsically over the pooled evaluation utterances (§7.1). C2 and C2q are evaluated *reference-fed*, taking annotated events and question roots as input to separate association and linking from extraction errors (§7.2, §7.3). C3 context policies are compared at budgets 64, 100, 128, and 256 (§7.4). The end-to-end error analysis crosses C1 input (reference vs. predicted) with context (*full*, no eviction, vs. *bounded* at budget 64), and the integrated run executes the full pipeline at budget 64 (§7.5).

Runs and statistical protocol. Each reported result comes from a single run with fixed seeds. Routing and context-policy comparisons use conversation-cluster bootstrap CIs and paired significance tests; the full protocol is in Appendices L and M.

C3 Experimental setup. The C3 pairwise MLP is trained with mean squared error using 23,409 source–target pairs, with conversation-level train/validation splitting. We evaluate on 120 held-out test conversations with C1 fixed, comparing the learned policy against FIFO, sliding-window, random, attention-based eviction, and an unlimited-buffer condition under $B \in \{64, 100, 128, 256\}$. Full training and implementation details are provided in Appendix J.

Metrics. Definitions are in Appendix I.

7 Results

Across the pipeline, selectivity creates an accuracy–cost trade-off: confidence routing reduces verifier

use while preserving competitive exact-link accuracy, and learned context selection is strongest under tight budgets.

7.1 C1: zero-shot to fine-tuned extraction

Zero-shot prompting is not a viable extractor: macro-F1 is only 0.212 (7-class), with the worst failures on the classes the tracker depends on most. Fine-tuning closes most of this gap, reaching 0.717 (training trajectory in Table 11 and Figure 10, appendix). On correctly detected events, the fine-tuned model’s one-sentence summaries score ROUGE-1 (Lin, 2004) 0.662 and BERTScore F1 0.862 on machine-label references ($n=449$).

7.2 Agenda opening and association (C2)

To isolate C2, we replace C1 outputs with reference model-facing events. On 4,190 child events with a non-null [delete: gold]annotated parent across 207 conversations, exact parent-ID accuracy is 0.896 (3,754/4,190)[delete: , up from 0.837 (3,507/4,190) before the tracker update], compared with 0.837 (3,507/4,190) for a base tracker without duplicate-root merging, topic-only semantic parent search, and active-agenda fallback.

[delete: The update jointly adds duplicate-root merging, topic-only semantic parent search, and active-agenda fallback; we do not attribute the gain to any one change. It reduces]The three mechanisms were introduced jointly, so we do not attribute the gain to any one of them. Together they reduce non-null parent errors from 683 to 436, but [delete: gold-null abstention falls from 0.515 to 0.415 because over-linking increases. Across all 5,922 child events, accuracy changes from 0.743 to 0.755]the tracker abstains less often on children that have no annotated parent. Across all 5,922 child events, accuracy therefore improves only from 0.743 to 0.755. Derived resolution states reach 0.903 balanced accuracy against manual labels (state definitions in §4.2). Extended C2 analyses appear in Appendix K. Separately, the predicted-C1 integrated run recovers 193 of 403 [delete: gold]annotated agendas, including 175 with zero opening delay. Of 2,090 child events, 1,954 arrive after their [delete: gold]annotated parent is present and 1,596 are linked exactly (Figure 9, appendix). These figures diagnose end-to-end extraction and association, not reference-fed C2 performance.

Table 1: Held-out C2q routing comparison on 140 non-training conversations (distance-order Top-2, fixed band (0.10, 0.75)).

Route	Bal. acc.	Exact ID	Ans. rec.	Unans. rec.	Calls
<i>Patient questions (n = 95; 6 unanswered)</i>					
NLI only	.737	.640	.640	.833	0
Verifier only	.755	.798	.843	.667	118
Length routing	.803	.607	.607	1.000	19
Confidence routing	.727	.775	.787	.667	20
<i>Physician questions (n = 1,219; 16 unanswered)</i>					
NLI only	.655	.695	.747	.563	0
Verifier only	.680	.796	.860	.500	1,475
Length routing	.704	.728	.782	.625	542
Confidence routing	.683	.812	.866	.500	255

7.3 C2q: exact linking and answerability

Table 1 reports the frozen four-way comparison on the 140 non-training conversations. The C1-v3 verifier was trained on the excluded 67 training conversations; NLI scores remain encounter-family cross-fitted. §8 examines the verifier’s contribution.

Length routing dispatches each candidate by reply form rather than by confidence: a reply that normalizes to one to eight words, contains at least one marker from a fixed affirmative, negative, or auxiliary list, and is not a bare backchannel is sent to the C1-v3 verifier, and every other reply is accepted when its NLI score reaches τ_A . The first accepted candidate in distance order is linked, and the question is reported unanswered when both Top-2 candidates are rejected. Length and confidence routing therefore share the same judges and differ only in the dispatch rule, by reply form or by NLI confidence band.

Lookahead. Most answers occur within one utterance after the question. Increasing the physician lookahead from 2 to 4 captures answers to 28 additional questions. [delete: Wider windows add almost no further answers.] Wider windows add few further physician answers. A full-transcript review of the questions unanswered at $K = 4$ shows most patient ones are eventually answered, some only tens of utterances later (Figure 13, appendix), so the window trades this late coverage for a small, high-precision candidate set. We therefore use $K = 4$ as the elbow (answer-position rates in Appendix L).

On the held-out split, length routing gives the highest balanced accuracy for both roles (0.803 patient; 0.704 physician). Confidence routing reaches 0.775 patient and 0.812 physician exact-ID accuracy while reducing verifier calls by about 83% relative to verifier-only (118 to 20; 1,475 to 255). It gives the best physician exact-ID accuracy, whereas verifier-only gives the best patient exact-ID accu-

racy (0.798). Length routing attains its balanced-accuracy lead by abstaining aggressively, and its exact-ID accuracy falls to 0.607 for patient and 0.728 for physician questions. Exact, verifiable answer links are the target output of the task (§3), so we adopt confidence routing as the primary operating point: it gives the best physician and near-best patient exact-ID accuracy at 17–20% of the verifier-only calls. The held-out statistics support this choice: the length-routing balanced-accuracy lead is not significant (paired difference CIs include zero for both roles), confidence routing is significantly more often correct per question than length routing (Holm-adjusted $p = 0.0079$ patient, 2.4×10^{-15} physician), its exact-ID gap to verifier-only is not significant in either role, and its exact-ID advantage over NLI-only is significant in both (Appendix L). The Top-1/Top-2 and threshold sweeps below use the full corpus and are retained only as diagnostics; because they include the verifier’s 67 training conversations, they are not used for [delete: headline test performance]the primary test results.

Sensitivity to answer phrasing. A single-human-reviewed pilot rephrases the same answers from short to long while holding the questions, candidate order, and reference links fixed (10 paired cases, 5 answered and 5 unanswered, over 20 synthetic dialogues). Averaged over five retrained NLI scorers, length routing drops from 1.000 to 0.700 balanced accuracy on the long variants (per-scorer drops 0.200 to 0.400): no reply qualifies as short, its verifier dispatches fall from 11 to zero, and the route degenerates to NLI-only. Confidence routing moves 0.040 (0.740 to 0.700), within its per-scorer range, and answered-side exact-ID stays at or near 1.000 for both routes. The fragile part is the dispatch rule rather than the judges: length routing keys on surface form, so rephrasing alone removes its verifier route, while the confidence band is indifferent to phrasing length. An N=10 pilot supports no generalization claim, but it cautions against reading the balanced-accuracy lead of length routing in Table 1 as robust to how answers happen to be phrased.

Retention, band selection, and sensitivity. Top-2 retention trades more verifier calls for exact-link accuracy, and validation-selected bands do not consistently improve on the fixed band (full results in Appendix L). The threshold-band sweep (Figure 4) shows that answerability, exact linking, and verifier cost trade off rather than improve together, so we keep the fixed band.

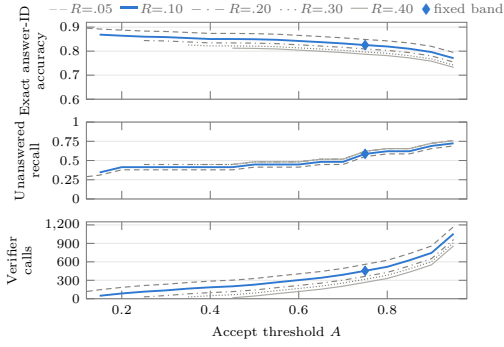


Figure 4: C2q threshold-band sensitivity (physician). Diamonds mark the fixed band (0.10, 0.75).

Table 2: Comparison of learned context selection and no eviction. Sum.Avg denotes the mean of the four summary-similarity metrics, and TypeF1.Avg denotes the mean of the five detection-oriented F1 metrics.

Setting	Tokens	Sum.Avg	AgendaC	TypeF1.Avg
Learned-64	54.8	0.6784	0.5186	0.6301
Learned-256	247.5	0.6852	0.4809	0.6166
No eviction	413.5	0.6833	0.4787	0.6180

7.4 C3 context-policy comparison

Additional C3 training and evaluation details and complete results are provided in Appendix M and Table 8.

Unlimited context is not an upper bound. The no-eviction condition retained an average of 413.5 tokens but achieved the lowest Agenda Completeness across all 21 conditions (0.4787), and its average summary similarity was 0.0019 below the learned policy at $B = 256$. In addition, its Agenda Item F1, Detail F1, Question F1, and Macro F1 all fell below every budget-constrained strategy. This result shows that retaining all accumulated summaries does not guarantee better performance than selective context retention.

Learned selection is most effective at the tightest context budgets. At $B = 64$ the learned policy achieved the highest score on six of the ten primary metrics, including all four summary-similarity metrics, Agenda Completeness, and Detail F1 (Table 8). Although heuristic policies remain stronger on several detection metrics, the learned policy provides the broadest overall advantage under the most restrictive context budget.

Larger contexts improve summary similarity but weaken agenda detection. As the context budget increased from 64 to 256 tokens, all four

Table 3: End-to-end error analysis on 140 held-out conversations; status macro-F1 is diagnostic only (column definitions and caveats in Appendix R).

Condition	Status	Link	First	Pat.	Phys.
Reference C1 (full = bounded)	0.276	0.789	0.938	0.727	0.683
Predicted C1, full	0.152	0.309	0.575	0.341	0.422
Predicted C1, bounded	0.148	0.352	0.647	0.419	0.483

summary-similarity metrics improved for every strategy. In contrast, Agenda Item F1 and the aggregate of the six agenda- and detection-oriented metrics declined consistently across all strategies. These results indicate that additional context improves summary similarity but can introduce interference in agenda detection.

Sliding achieves greater context efficiency than FIFO at large budgets. At $B = 256$, Sliding uses 33.37% fewer final-buffer tokens than FIFO while outperforming it on six of the ten primary metrics. Sliding leads on Agenda Completeness and all five detection-oriented F1 scores, whereas FIFO remains slightly stronger on the four summary-similarity metrics. Thus, which summaries are retained matters more than filling the available budget.

7.5 End-to-end agenda status

Table 3 reports the end-to-end error analysis. **Status macro-F1 is computed on the strict 144-node subset whose reference status is determinate under every condition, so it is diagnostic rather than a corpus-level result (Appendix R).** The reference-fed full and bounded rows are identical by construction (the reference-fed mock extractor does not condition on the buffer). Two findings stand out. First, swapping reference C1 events for predicted ones collapses parent linking and status F1; the attribution analysis (§9) traces these errors chiefly to missed agenda openings. Second, under predicted C1 the bounded 64-token context improves four of five metrics over the full buffer despite similar status macro-F1, suggesting that context selection filters extractor noise.

An integrated run executes the full $C1 \rightarrow C2 \rightarrow C3$ pipeline over all 207 conversations at budget 64 (C2q excluded, distinct from the intrinsic comparison of §7.4). The learned policy attains the highest detection macro-F1 of six context strategies (0.465 vs. 0.462–0.464) with a mean final buffer of 54.7 tokens against 374.2 unbounded, an 85.4% reduction at comparable performance. In

sum, tracking and linking are reliable once an agenda is opened, selective verification [delete: and bounded context selection outperform their indiscriminate alternatives]recovers most of the accuracy of always verifying at a small fraction of its cost, bounded context selection matches or beats retaining far more context, and the remaining limit is upstream in extraction recall.

8 Ablation and Analysis

Verifier routing and candidate ordering. Table 1 (§7.3) reports the held-out four-way comparison. Every [delete: gold]reference answer is present in the distance-order Top-2 candidate sets (89/89 patient; 1,203/1,203 physician), so candidate coverage is not the bottleneck on this split. Confidence routing misses 19 answered patient questions and 161 answered physician questions; among questions linked as answered, 1 patient and 65 physician links select the wrong utterance ID. It also falsely links 2 of 6 unanswered patient questions and 8 of 16 unanswered physician questions. The remaining errors therefore arise from routing, answerhood, and exact-utterance selection rather than candidate retrieval. For diagnosis only, the full-corpus route audit retains an archived HGB edge-score ordering comparison and the human-reviewed taxonomy of 43 unanswered rows (Appendix N). These analyses include the verifier’s training conversations and are not used for head-line performance.

Hindsight versus attention supervision in C3. On a 120-conversation test set with C1 fixed (atlas-c1-v3-llama-3.2-3b), we compare BERT-396, supervised by binary BERTScore-0.75 hindsight labels, with Attn-780, supervised by within-timestep attention percentile ranks. Attn-780 achieves higher scores on all ten primary metrics at budgets 64, 100, and 128, and retains an advantage on eight metrics at budget 256. This consistent advantage under tighter budgets suggests that semantic similarity to future summaries is an incomplete proxy for identifying the context actually utilized by C1. Detailed results at representative budgets are provided in Appendix O.

Answer-removal analysis. As a sensitivity analysis of standalone LLM verification, GPT-5.5 and Claude Opus 4.8 are evaluated as sole selectors. Both find the reference answer when it is present (97.4–97.7%), but after the answer utterance is re-

moved they still select another utterance in 16–19% of physician windows, consistent with prior abstention evaluations (Madhusudhan et al., 2025; Lavi et al., 2026) (Table 10, Appendix Q). The altered windows are not independently re-annotated, so the removal counts are a sensitivity probe rather than new unanswerable annotations. A stronger verifier alone does not solve abstention, supporting verification only in the uncertain band.

9 Discussion

Online clinical agenda tracking is a structured next-state prediction problem, and the end-to-end analysis (§7.5) shows where it fails: with reference extraction, 79% of status errors trace to answer linking; with the real extractor, 81–84% trace to missed agenda openings, making opening recall the largest single lever.

Selective verification and selective context. On the held-out split, no single route dominates every endpoint: length routing gives the highest balanced accuracy for both roles, verifier-only gives the highest patient exact-ID accuracy, and confidence routing gives the highest physician exact-ID accuracy while using about 83% fewer verifier calls than verifier-only (§7.3). The confidence band therefore provides an accuracy–cost operating point rather than a universal accuracy optimum, and a stronger verifier alone is not a substitute for abstention (§8). Context selection shows the same pattern: more context does not help (§7.4), and under a noisy extractor bounded selection beats the full buffer (§7.5). Summary quality and agenda coverage separate across policies, motivating selection signals that represent agenda structure explicitly (§8).

Patient–provider conversation dynamics. Our annotations quantify patterns the clinical-communication literature has described qualitatively, and the failure modes are asymmetric: when a physician’s question fails, the patient has usually *tried* to answer with a reply that is topically adjacent but off-target (15 of 29 reviewed physician non-answers, versus 3 of 14 on the patient side); when a patient’s question fails, the physician has usually *not* tried, deflecting rather than answering. Physician questions largely resolve themselves, so an agenda tracker is most useful for protecting the patient’s side of the conversation, where reliable abstention matters most. The full analysis and caveats appear in

10 Conclusion

ATLAS models clinical agenda understanding as a streaming transition over a persistent structured state: it extracts typed events, associates them with dynamic agendas, links open questions to exact answer utterances with calibrated abstention, and preserves condensed evidence under a fixed budget. [On the held-out 140-conversation split, confidence routing reaches 0.775 patient and 0.812 physician exact-link accuracy while reducing verifier calls by about 83% relative to verifier-only, and bounded context selection matches or beats retaining far more context.](#) Live agenda tracking with exact, verifiable answer links is therefore feasible with [\[delete: small on-device models; the\]small open-weight models](#). The remaining bottleneck is agenda-opening recall. Future work is better opening supervision, broader unanswered-question annotation, and real implementations on live recorded encounters.

Limitations

Real clinical conversations are rarely recorded and difficult to share ([Yim et al., 2023](#)), and MTS-Dialog provides dialogue-note pairs rather than question-answer linkage annotations ([Ben Abacha et al., 2023](#)). Our unanswered-question set covers this gap but is thin (14 patient, 29 physician), and extending it to real recordings is future work.

Several evaluation caveats remain. The end-to-end status F1 is an error-propagation analysis, not a primary benchmark, since only 144 of the 264 strict-pool agenda nodes are judgeable under all four conditions (Appendix R). C2q searches only the next four utterances and misses later answers. Attention-derived labels measure utilization, not causal effect. Summary quality is scored against machine-generated references, and significance testing covers the routing and context-policy comparisons but not the remaining component metrics.

All experiments use one ACI-BENCH-derived corpus of primary-care encounters, and generalization across specialties, institutions, and visit types is untested. In the DialogSum ([Chen et al., 2021b](#)) study (Appendix C), fine-tuned C1 keeps its advantage on all mapped classes, but agenda-opening recall is 0.05 at precision near 0.8. We do not claim cross-domain generalization, and a new domain needs its own opening supervision.

Finally, C1 depends on utterance-level multi-label annotations that are expensive to validate. The synthetic-data sweep plateaus at 100 conversations (Figure 7) without isolating whether annotation noise, data composition, or optimization is responsible, which motivates human-validated labels.

References

- Sean Allgood, Jenny Park, Kiana Soleiman, Somnath Saha, Dingfen Han, Amanda McArthur, Richard D. Moore, and Mary Catherine Beach. 2023. Taxonomy and effectiveness of clinician agenda-setting questions in routine ambulatory encounters: A mixed method study. *Patient Education and Counseling*, 115:107889.
- Sotiris Anagnostidis, Dario Pavllo, Luca Biggio, Lorenzo Noci, Aurelien Lucchi, and Thomas Hofmann. 2023. [Dynamic context pruning for efficient and interpretable autoregressive transformers](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Laszlo A. Belady. 1966. A study of replacement algorithms for a virtual-storage computer. *IBM Systems Journal*, 5(2):78–101.
- Asma Ben Abacha, Wen-wai Yim, Yadan Fan, and Thomas Lin. 2023. [An empirical study of clinical note generation from doctor-patient encounters](#). In *Proceedings of EACL*, pages 2291–2302.
- Aydar Bulatov, Yuri Kuratov, and Mikhail S. Burtsev. 2022. [Recurrent memory transformer](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Rafael Carranza and Mateo Alejandro Rojas. 2025. Interpretable and robust dialogue state tracking via natural language summarization with LLMs. *arXiv preprint arXiv:2503.08857*.
- Howard Chen, Ramakanth Pasunuru, Jason Weston, and Asli Celikyilmaz. 2023. Walking down the memory maze: Beyond context limit through interactive reading. *arXiv preprint arXiv:2310.05029*.
- Jifan Chen, Eunsol Choi, and Greg Durrett. 2021a. [Can NLI models verify QA systems’ predictions?](#) In *Findings of EMNLP*, pages 3841–3854.
- Yulong Chen, Yang Liu, Liang Chen, and Yue Zhang. 2021b. [DialogSum: A real-life scenario dialogue summarization dataset](#). In *Findings of ACL*, pages 5062–5074.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. Mem0: Building production-ready AI agents with scalable long-term memory. In *Proceedings of ECAI*, pages 2993–3000.
- Eunsol Choi, He He, Mohit Iyyer, Mark Yatskar, Wentau Yih, Yejin Choi, Percy Liang, and Luke Zettlemoyer. 2018. [QuAC: Question answering in context](#). In *Proceedings of EMNLP*.

- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. [QLoRA: Efficient fine-tuning of quantized LLMs](#). In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 10088–10115.
- Yujie Feng, Zexin Lu, Bo Liu, Liming Zhan, and Xiaoming Wu. 2023. Towards LLM-driven dialogue state tracking. In *Proceedings of EMNLP*, pages 739–755.
- Etienne Fortier-Dubois and Domenic Rosati. 2023. Using contradictions improves question answering systems. In *Proceedings of ACL*, pages 827–840.
- Logan Golia and Jugal Kalita. 2023. Action-item-driven summarization of long meeting transcripts. In *Proceedings of NLP4IR*, pages 91–98.
- Tyler Haberle, Courtney Cleveland, Greg L. Snow, Chris Barber, Nikki Stookey, Cari Thornock, Laurie Younger, Buzzy Mullahkhel, and Diego Ize-Ludlow. 2024. The impact of nuance DAX ambient listening AI documentation: A cohort study. *Journal of the American Medical Informatics Association*, 31(4):975–979.
- Huiqiang Jiang, Qianhui Wu, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2023. [LLMLingua: Compressing prompts for accelerated inference of large language models](#). In *Proceedings of EMNLP*.
- Naga Sasidhar Kanaparth, Yenny Villuendas-Rey, Tolupe Bakare, Zihan Diao, Mark Iscoe, Andrew Loza, Donald Wright, Conrad Safranek, Isaac V. Faustino, Alexandria Brackett, Edward R. Melnick, and R. Andrew Taylor. 2025. [Real-world evidence synthesis of digital scribes using ambient listening and generative artificial intelligence for clinician documentation workflows: Rapid review](#). *JMIR AI*, 4:e76743.
- Andreas Kathol and Gokhan Tur. 2008. [Extracting question/answer pairs in multi-party meetings](#). In *Proceedings of ICASSP*, pages 5053–5056.
- Maor Juliet Lavi, Tova Milo, and Mor Geva. 2026. [Detecting \(un\)answerability in large language models with linear directions](#). In *Proceedings of EACL*, pages 682–699.
- Yizhan Li, Sifan Wu, Christopher Smith, Thomas Lo, and Bang Liu. 2025. Improving clinical note generation from complex doctor-patient conversation. In *Advances in Knowledge Discovery and Data Mining (PAKDD)*, pages 209–221.
- Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. 2024a. [SnapKV: LLM knows what you are looking for before generation](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Zekun Li, Zhiyu Zoey Chen, Mike Ross, Patrick Huber, Seungwhan Moon, Zhaojiang Lin, Xin Luna Dong, Adithya Sagar, Xifeng Yan, and Paul A. Crook. 2024b. Large language models as zero-shot dialogue state tracker through function calling. In *Proceedings of ACL*, pages 8688–8704.
- Chin-Yew Lin. 2004. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81.
- Jiaqing Liu, Chong Deng, Qinglin Zhang, Qian Chen, and Wen Wang. 2023a. Meeting action item detection with regularized context modeling. In *Proceedings of ICASSP*, pages 1–5.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. [Lost in the middle: How language models use long contexts](#). *Transactions of the Association for Computational Linguistics*, 12.
- Xinyan Liu, Jiaxin Lu, and Ang Xu. 2025. Dynamic agenda-aware real-time meeting summarization with large language models. *Journal of King Saud University – Computer and Information Sciences*, 37(9):275.
- Zichang Liu, Aditya Desai, Fangshuo Liao, Weitao Wang, Victor Xie, Zhaozhao Xu, Anastasios Kyrilidis, and Anshumali Shrivastava. 2023b. [Scissorhands: Exploiting the persistence of importance hypothesis for LLM KV cache compression at test time](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Junru Lu, Siyu An, Mingbao Lin, Gabriele Pergola, Yulan He, Di Yin, Xing Sun, and Yunsheng Wu. 2023. [MemoChat: Tuning LLMs to use memos for consistent long-range open-domain conversation](#). *arXiv preprint arXiv:2308.08239*.
- Mengxian Lyu, Cheng Peng, Xiaohan Li, Patrick Balian, Jiang Bian, and Yonghui Wu. 2025. Automatic summarization of doctor-patient encounter dialogues using large language model through prompt tuning. *AMIA Joint Summits on Translational Science Proceedings*, 2025:342–349.
- Nishanth Madhusudhan, Sathwik Tejaswi Madhusudhan, Vikas Yadav, and Masoud Hashemi. 2025. [Do LLMs know when to NOT answer? investigating abstention abilities of large language models](#). In *Proceedings of COLING*, pages 9329–9345.
- M. Kim Marvel, Ronald M. Epstein, Kristine Flowers, and Howard B. Beckman. 1999. Soliciting the patient’s agenda: Have we improved? *JAMA*, 281(3):283–287.
- Meta AI. 2024. Llama 3.2 model card: Llama-3.2-3b. <https://huggingface.co/meta-llama/Llama-3.2-3B>.
- Amirkeivan Mohtashami and Martin Jaggi. 2023. [Random-access infinite context length for transformers](#). In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 54567–54585.
- Luca Moschella, Laura Manduchi, and Ozan Sener. 2026. Learning to evict from key-value cache. In *Proceedings of ICML*.

- Jesse Mu, Xiang Lisa Li, and Noah Goodman. 2023. [Learning to compress prompts with gist tokens](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Mark Neumann, Daniel King, Iz Beltagy, and Waleed Ammar. 2019. ScispaCy: Fast and robust models for biomedical natural language processing. In *Proceedings of the BioNLP Workshop*, pages 319–327.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2023. [MemGPT: Towards LLMs as operating systems](#). *arXiv preprint arXiv:2310.08560*.
- Alex Papadopoulos Korfiatis, Francesco Moramarco, Radmila Sarac, and Aleksandar Savkov. 2022. [Pri-Mock57: A dataset of primary care mock consultations](#). In *Proceedings of ACL*, pages 588–598.
- Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, Chloe Hillier, and Timothy P. Lillicrap. 2020. [Compressive transformers for long-range sequence modelling](#). In *International Conference on Learning Representations (ICLR)*.
- Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. [Know what you don’t know: Unanswerable questions for SQuAD](#). In *Proceedings of ACL*, pages 784–789.
- Vijay Janapa Reddi. 2025. [Generative AI at the edge: Challenges and opportunities](#). *ACM Queue*, 23(2):79–137.
- Siva Reddy, Danqi Chen, and Christopher D. Manning. 2019. [CoQA: A conversational question answering challenge](#). *Transactions of the Association for Computational Linguistics*, 7.
- Ailyn Sierpe, Renata W. Yen, Gabrielle Stevens, Aricca D. Van Citters, Glyn Elwyn, and Catherine H. Saunders. 2024. Agenda-setting in the clinical encounter: A systematic review protocol. *PLoS ONE*, 19(10):e0312613.
- Naykky Singh Ospina, Kari A. Phillips, Rene Rodriguez-Gutierrez, Ana Castaneda-Guarderas, Michael R. Gionfriddo, Megan E. Branda, and Victor M. Montori. 2019. [Eliciting the patient’s agenda: Secondary analysis of recorded clinical encounters](#). *Journal of General Internal Medicine*, 34:36–40.
- Andreas Stolcke, Klaus Ries, Noah Coccaro, Elizabeth Shriberg, Rebecca Bates, Daniel Jurafsky, Paul Taylor, Rachel Martin, Carol Van Ess-Dykema, and Marie Meteer. 2000. Dialogue act modeling for automatic tagging and recognition of conversational speech. *Computational Linguistics*, 26(3):339–373.
- Aaron A. Tierney, Gregg Gayre, Brian Hoberman, Britt Mattern, Manuel Ballesca, Patricia Kipnis, Vincent Liu, and Kristine Lee. 2024. Ambient artificial intelligence scribes to alleviate the burden of clinical documentation. *NEJM Catalyst Innovations in Care Delivery*, 5(3).
- Bing Wang, Xinnian Liang, Jian Yang, Hui Huang, Zhenhe Wu, Shuangzhi Wu, Zejun Ma, and Zhoujun Li. 2026. SCM: Enhancing large language model with self-controlled memory framework. In *Database Systems for Advanced Applications (DASFAA)*, pages 188–203.
- Junda Wang, Zonghai Yao, Zhichao Yang, Huixue Zhou, Rumeng Li, Xun Wang, Yucheng Xu, and Hong Yu. 2024. [NoteChat: A dataset of synthetic patient-physician conversations conditioned on clinical notes](#). In *Findings of ACL*, pages 15183–15201.
- Qingyue Wang, Yanhe Fu, Yanan Cao, Shuai Wang, Zhiliang Tian, and Liang Ding. 2025. Recursively summarizing enables long-term dialogue memory in large language models. *Neurocomputing*, 639:130193.
- Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. 2020. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems*, volume 33, pages 5776–5788.
- Yu Wang and Xi Chen. 2025. MIRIX: Multi-agent memory system for LLM-based agents. *arXiv preprint arXiv:2507.07957*.
- Adina Williams, Nikita Nangia, and Samuel R. Bowman. 2018. [A broad-coverage challenge corpus for sentence understanding through inference](#). In *Proceedings of NAACL*.
- Yuhuai Wu, Markus N. Rabe, DeLesley Hutchins, and Christian Szegedy. 2022. [Memorizing transformers](#). In *International Conference on Learning Representations (ICLR)*.
- Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. 2024. [Efficient streaming language models with attention sinks](#). In *International Conference on Learning Representations (ICLR)*.
- Wen-wai Yim, Yujuan Fu, Asma Ben Abacha, Neal Snider, Thomas Lin, and Meliha Yetisgen. 2023. [Acibench: a novel ambient clinical intelligence dataset for benchmarking automatic visit note generation](#). *Scientific Data*, 10:586.
- Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. 2023. [H2O: Heavy-hitter oracle for efficient generative inference of large language models](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.

Disclosure of AI Use

Large language model tools assisted with editing and manuscript preparation. AI systems used in the pipeline and evaluation are documented where they appear. The authors remain fully responsible for all content.

A Code and Data Availability

All source code (the ATLAS pipeline, training and evaluation scripts, and the annotation tooling) and all data introduced in this paper (the 207-conversation annotated corpus, the synthetic conversations, and the DialogSum conversion) will be made publicly available upon acceptance, under a license that allows free use for research purposes.

B Extended Related Work

Clinical dialogue resources and note generation.

Beyond the corpora cited in the main text, doctor–patient dialogue resources include mock consultations (Papadopoulos Korfiatis et al., 2022) and clinical summarization benchmarks (Ben Abacha et al., 2023); a taxonomy of clinician agenda-setting questions quantifies their behavioral effects in routine encounters (Allgood et al., 2023); agenda-aware meeting summarization tracks a *known* agenda (Liu et al., 2025). Deployment evidence for ambient documentation includes a large cohort study of a commercial system (Haberle et al., 2024). Note generation from complex clinical conversations is an active line, including prompt-tuned summarization of encounter dialogues (Lyu et al., 2025) and pipelines targeting complex multi-topic visits (Li et al., 2025).

Dialogue-state tracking and meeting understanding.

Natural-language state summarization offers an interpretable alternative to slot filling (Carranza and Rojas, 2025), and meeting-understanding work extracts action items from long transcripts, including with regularized context modeling (Golia and Kalita, 2023; Liu et al., 2023a). LLM-based trackers span function calling (Li et al., 2024b) and instruction-driven prompting (Feng et al., 2023). Relative to slot filling, open-world agenda construction is closer to incremental event extraction and coreference: the state must support creation, revisiting, and merging over the whole encounter, and its status semantics are derived deterministically from its contents rather than predicted.

Answerability and verification. Conversational QA with unanswerable questions is benchmarked in CoQA (Reddy et al., 2019); contradiction signals have been used to improve QA reliability (Fortier-Dubois and Rosati, 2023); and dialogue-act modeling (Stolcke et al., 2000) underlies early work on question–answer pair extraction from multi-party meeting speech, including non-adjacent pairs

(Kathol and Tur, 2008). Unlike one-shot answerability against a fixed passage, a standing question in a live encounter may be resolved several turns after it is asked, and the exact-utterance requirement further distinguishes answer *linking* from binary answerability classification.

Context compression and agent memory. KV-cache compression spans streaming attention sinks (Xiao et al., 2024), heavy-hitter retention (Zhang et al., 2023), and learned eviction (Moschella et al., 2026); further techniques include landmark tokens (Mohtashami and Jaggi, 2023), importance-persistence-based retention (Liu et al., 2023b), dynamic context pruning (Anagnostidis et al., 2023), and observation-window selection (Li et al., 2024a); gist tokens compress prompts into reusable prefixes (Mu et al., 2023). Memory-augmented architectures include compressive transformers and recursive summarization (Rae et al., 2020; Wang et al., 2025), k NN-augmented memorizing transformers (Wu et al., 2022), recurrent memory transformers (Bulatov et al., 2022), and interactive-reading agents that navigate a memory tree (Chen et al., 2023). LLM-agent memory systems beyond those cited in the main text include self-controlled memory (Wang et al., 2026), memo-structured chat memory (Lu et al., 2023), production-oriented scalable long-term memory (Chhikara et al., 2025), and multi-agent memory (Wang and Chen, 2025). C3’s selection problem echoes classical cache replacement, with “future utility” as task-specific state preservation rather than access frequency (Belady, 1966).

C Cross-Domain Feasibility Study Details

To examine whether the agenda-centered annotation contract is specific to clinical conversations, we conducted an independent feasibility study on DialogSum: 1,611 prepared two-party conversations (filtered from 13,460) were converted to the ATLAS annotation format under a generalized seven-type schema and annotated with a standalone pipeline. On 15,162 utterances with machine reference labels, the fine-tuned C1 retains its gains over zero-shot across all mapped classes, indicating that the model learned utterance structure rather than hospital-specific content. Agenda-item detection, however, transfers asymmetrically: recall is only 0.05 while precision remains near 0.8 (the model abstains rather than guesses when clinical topic-opening cues are absent from everyday di-

alogue), and the agendas that *are* detected retain acceptable summary quality. The cross-domain bottleneck is therefore detection, not summarization. This study is excluded from the main human-annotated denominator and we do not claim successful cross-domain generalization; it shows the annotation contract is operable out of domain and isolates detection recall as the transfer gap.

Data preparation. Starting from 13,460 unique DialogSum dialogues, we normalized the raw CSV structure with duplicated reference summaries, retained only two-party conversations, and removed duplicate conversations and clearly off-domain topics. We then filtered for agenda-bearing interactions in business, office, banking, finance, and service-request settings, yielding 1,611 prepared conversations (1,470 train / 47 validation / 94 test).

Schema mapping. A standalone cross_domain toolkit, separate from the clinical ATLAS codebase, converts DialogSum conversations into the ATLAS-compatible relevant + detail_list format. Clinical-domain-specific labels and prompt wording were removed and annotation types suitable for general dialogue were defined: agenda_item, detail, action_item, background_context, follow_up, question, and question_unanswered. Encounter-specific categories (medication, social_history) were removed; action_item and background_context capture task execution, decisions, commitments, and contextual information. The core agenda-centered structure (agenda boundaries, supporting details, follow-ups, questions, and unanswered questions) is preserved to maintain compatibility with the ATLAS annotation contract.

Scope classification. Deterministic scope classification labels 1,351 of the 1,611 conversations as agenda-bearing, 239 as ambiguous, and 21 as primarily social dialogue, so that social or off-scope conversations can be excluded from, or analyzed separately in, annotation-performance evaluation.

Annotation run. Using the general-v1-agenda-r4 prompt with gemini-3.1-flash-lite, all 1,611 prepared conversations were annotated, producing 1,611 annotation JSON files conforming to the general-v1 schema. The pool contains 1,710 agenda items, 5,166 details, 2,527 questions, 925 action items, 90 follow-ups, 59 background contexts, and 4 unanswered questions. Figure 5 reports per-class detection F1 and agenda summary quality for

zero-shot versus fine-tuned C1.

D Supplementary Figures

This appendix collects the supplementary figures referenced from the main text: the V2→V3 per-class comparison (Figure 6), the synthetic-data saturation sweep (Figure 7), the full threshold-band sweep (Figure 8), C2 opening and linking (Figure 9), zero-shot C1 per-class performance (Figure 10), and the C1 hyperparameter sweep (Figure 11).

E Annotation Schema and Guidelines

Annotation proceeds in a two-stage teacher-validate loop (§5.1). In the *batch* stage the teacher model (Gemini 3.1 Flash Lite) receives the full transcript together with the reference clinical note and the chief-complaint metadata, and returns per-utterance structured annotations; in the *validate* stage a second pass samples [delete: ~5%]10% of annotated utterances and [delete: scores each on faithfulness (0–3), type correctness (0/1), and summary quality (0–3)]accepts or rejects each entry, recording the cause of every rejection.

Each annotated utterance carries a detail_list of one or more entries. Every entry specifies a *type* from the seven-label schema (agenda_item, question, detail, medication, social_history, follow_up, question_unanswered) and a one-sentence *summary* that resolves pronouns and implicit references; in evaluation mode an eval_only block adds the linked agenda identifier, first-mention flag, resolution status, and SOAP-aligned clinical category (§5.1). The reference clinical note is the primary annotation criterion: content documented in the note must be annotated, content absent from it may be excluded, and all clinically relevant questions are captured regardless of note coverage. A single utterance may carry multiple entries of *different* types (an agenda_item plus a question when a topic transition co-occurs with an inquiry), while multiple pieces of information of the same type merge into one summary. question_unanswered is reserved for questions that receive no substantive clinical response anywhere in the encounter[delete: ; this label never occurs naturally in the human corpus and is supplied by]. The human annotators never assign this utterance-level label. For human questions, unanswered status is instead derived from the

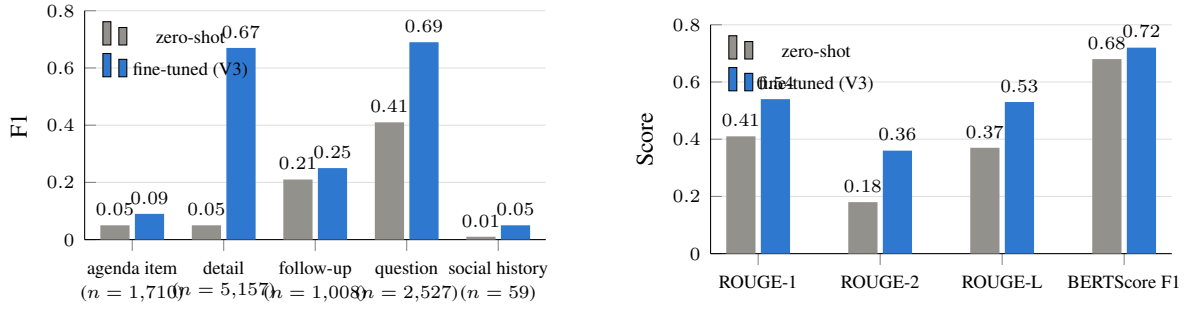


Figure 5: Cross-domain feasibility on DialogSum. **Left:** detection F1 (15,162 utterances, machine reference labels), zero-shot versus fine-tuned V3; class support is shown below each label. **Right:** agenda summary quality on rows detected as agendas by each model (zero-shot $n = 90$; fine-tuned V3 $n = 85$). Agenda-item detection recall is the transfer gap (0.05 at precision near 0.8).

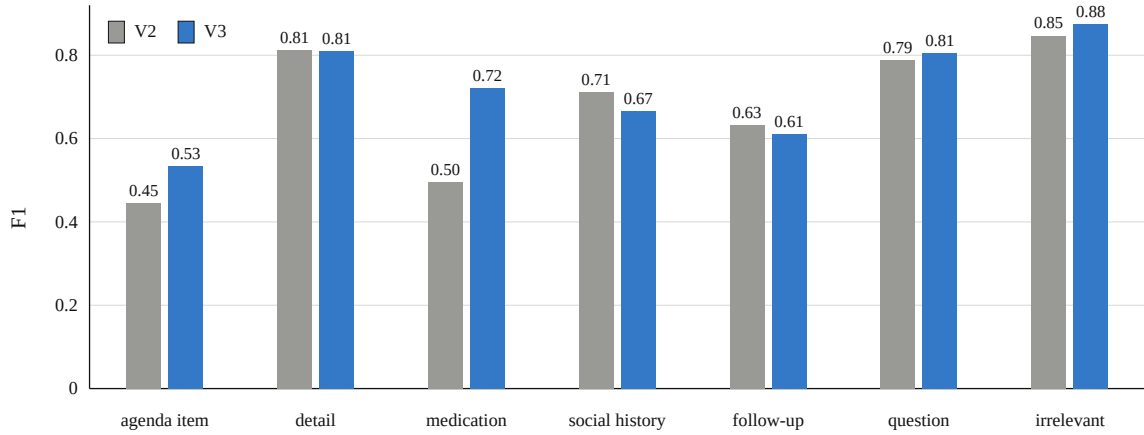


Figure 6: Per-class F1 for C1 V2 and V3 on the fixed 7,735-utterance evaluation pool. V3 raises 7-class macro-F1 from 0.676 to 0.717. The 7-class reporting protocol excludes question_unanswered.

evaluation-pass answer links: a question with no linked answer utterance within the frozen lookahead-4 window is labeled unanswered (the 43 questions of Appendix N). Utterance-level question_unanswered exemplars come only from the controlled synthetic generation of §5.2. [delete: Teacher annotations were validated by a second-pass sample (§5.1): over 88 sampled annotations, the acceptance rate is 88.6%, mean faithfulness 2.83/3, and type accuracy 95.5%, with rejected annotations corrected manually.]Teacher annotations were validated by a second-pass review of a 10% random sample (§5.1): of 1,197 sampled utterances, 1,103 (92.1%) were accepted, and the 94 rejected utterances carried 100 faulty entries, all corrected manually. Over the 100 entries, the rejection causes split into note-only content (32%), misinterpretation (30%), wrong-line reference (13%), speaker error (11%), type classification

(8%), and information omission (6%). The note determines which conversation content must be annotated and never supplies content itself. An annotation drawn from the note without support in the conversation is an error, the note-only category above. The full guideline document, per-type examples, and prompt templates ship with the dataset release.

Annotation agreement. Two annotators independently labeled five training conversations (222 utterances) with the utterance-level type scheme (six clinical types plus irrelevant, question_unanswered excluded). All-label exact match is 0.712 and macro-averaged Cohen’s κ over the seven labels is 0.651, with per-label κ from 0.873 (medication) to 0.374 (agenda_item). Most disagreements sit on the relevance boundary (26 of the 64 disagreeing utterances pit detail

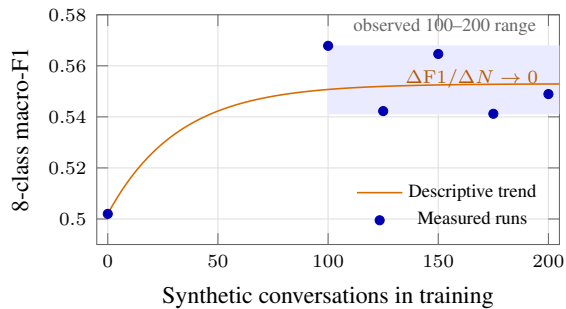


Figure 7: C1 synthetic-data scaling on the fixed 1,070-row validation set. Dots show measured 8-class macro-F1; the orange line is a descriptive saturation guide, not a fitted scaling law. The no-synthetic point is the existing C1-v2 reference, whereas 100–200 use cumulative C1-v3 synthetic data. Beyond 100 conversations, no run exceeds the 0.568 score at 100.

or question against irrelevant), and agreement is lowest on agenda openings, the same class the extractor finds hardest (§7.1). This is a pre-adjudication pilot on the train split. Separately, comparing the teacher annotations against the corrected human reference over all 207 conversations finds differences confined to 12 conversations (490 utterances). Within those conversations the differences are largely wholesale, consistent with source-transcript corrections applied after teacher annotation, such as speaker-role swaps that flip speaker-dependent labels, rather than distributed label disagreement. The remaining 195 conversations show no differences.

F Corpus Annotation Statistics and Synthetic Generation Details

Tables 4 and 5 report the annotation type distributions of the full 207-conversation ACI corpus and its 67-conversation training split. Beyond the counts reported in §5.1: the three held-out test sets contain 2,091 / 2,232 / 2,342 utterances; the curated corpus carries 6,325 detail-list entries; 58.5% of agenda nodes carry a follow-up; and multi-label utterances occur, up to a social-history event plus three agenda items in a single utterance. Table 4 reports the first-pass annotation distribution: 6,408 entries, including 423 agenda_item labels. The reported evaluations instead use the separate evaluation-pass annotations, which contain 403 agenda items, 2,261 physician questions, and 197 patient questions.

Unanswered-question generation. The constrained generation pipeline (§5.2) targets three

Table 4: Annotation type distribution of the 207 ACI conversations ($N_{\text{utt}}=11,478$, $N_{\text{ann}}=6,408$).

Label type	Count	%
detail	3 013	47.0
question	1 862	29.1
follow_up	557	8.7
agenda_item	423	6.6
social_history	338	5.3
medication	215	3.4
question_unanswered	0	0.0

Note: Percentages may not sum to 100% due to rounding.

Table 5: Annotation type distribution of the 67 ACI training conversations ($N_{\text{utt}}=3,743$, $N_{\text{ann}}=2,013$).

Label type	Count	%
detail	943	46.8
question	597	29.7
follow_up	170	8.4
agenda_item	131	6.5
social_history	109	5.4
medication	63	3.1
question_unanswered	0	0.0

implicit non-answer mechanisms, distributed approximately equally across the 100 synthetic conversations (5,478 utterances):

- ignore**: minimal filler acknowledgement followed by a new clinical question ($n=35$),
- counter-question**: an unrelated question replacing the answer ($n=33$), and
- topic-shift**: a pivot to a different recommendation ($n=32$).

Each conversation undergoes automated validation enforcing 44 forbidden deferral phrases, speaker-alternation repair, keyword-overlap checks confirming the question remains unanswered until the final turn, and positional constraints (target index sampled uniformly from the 20th–80th percentile of conversation length to prevent early clustering). Failed conversations are regenerated up to three times; IDs are assigned only upon success to guarantee contiguous numbering. The resulting conversations are annotated with the same batch prompt used for the human-annotated set, followed by a post-processing step that verifies the designed unanswered utterance carries the correct label. **These checks are structural: they verify surface constraints and label placement, not the clinical plausibility or semantic correctness of the generated dialogue, which no blinded clinical review has assessed.** Table 6 reports the resulting annotation distribution.

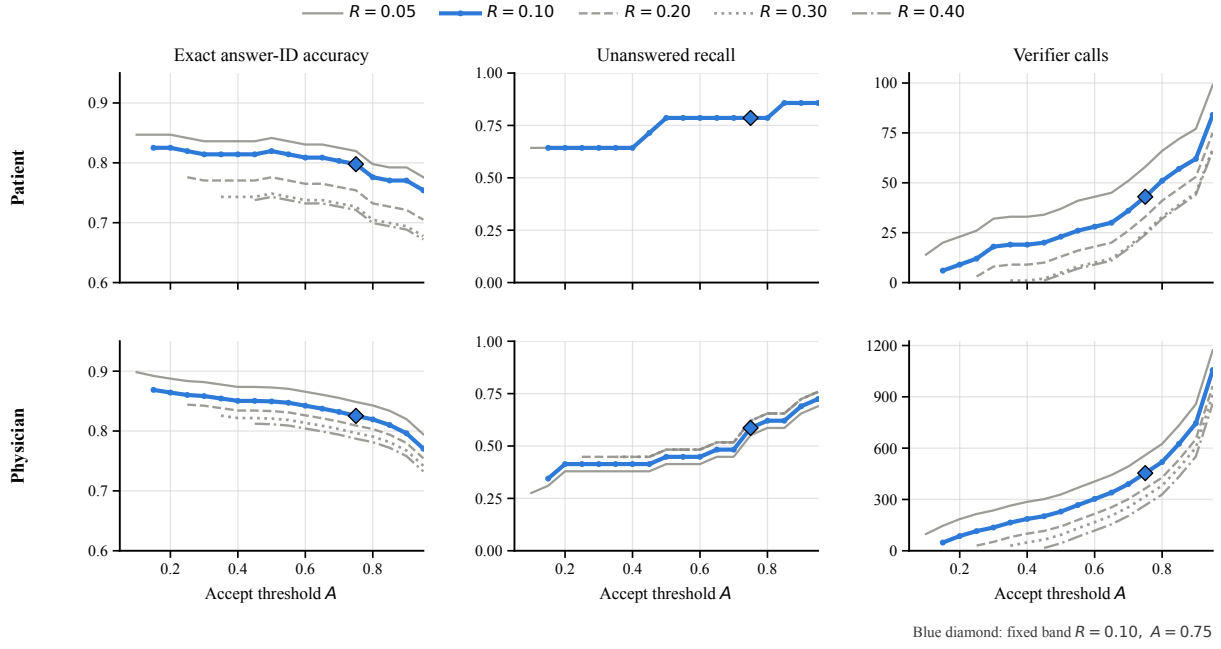


Figure 8: C2q threshold-band sensitivity with human-annotated question roots and distance-order Top-2 candidates, both roles (the physician row is excerpted as Figure 4 in the main text). Curves vary the accept threshold A for five reject thresholds R ; the blue curve highlights $R = 0.10$, and blue diamonds mark the fixed working band $(0.10, 0.75)$. The full sweep is a post-hoc sensitivity analysis, not a validated threshold-selection result.

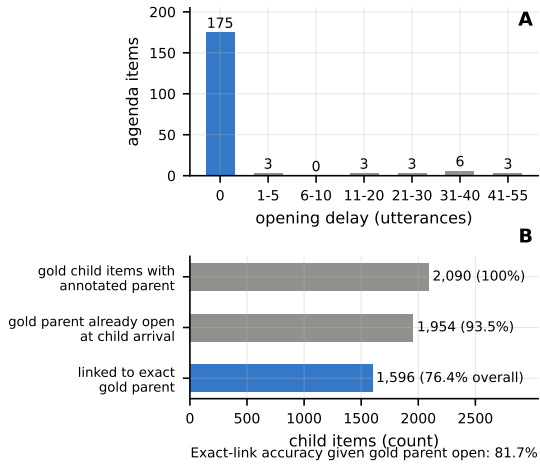


Figure 9: C2 agenda tracking. (A) Opening delay for the 193 annotated agendas the tracker opens: 175 (91%) open with zero delay; the failure mode is no opening, not late opening. (B) Linking over 2,090 annotated child items with a labeled parent: the annotated parent is open at child arrival for 1,954 (93.5%), and 1,596 are linked to it exactly (76.4% overall; 81.7% given parent open).

G Synthetic Unanswered-Question Data Quality

We compare the 100 synthetic unanswered-question conversations (ST) against the 67 real ACI-BENCH training conversation (D2N) to verify that the augmentation corpus is structurally and

Table 6: Annotation type distribution of the 100 synthetic unanswered-question conversations ($N_{\text{utt}}=5,478$, $N_{\text{ann}}=4,355$).

Label type	Count	%
question	1 776	40.8
detail	1 586	36.4
follow_up	290	6.7
agenda_item	286	6.6
social_history	191	4.4
question_unanswered	118	2.7
medication	108	2.5

Note: Percentages may not sum to 100% due to rounding.

clinically plausible.

Annotation distribution Figure 12(a) overlays ten annotation metrics on a radar plot. Annotation density is higher in synthetic data (0.754 vs. 0.475 annotations per utterance), partly because the generator is instructed to produce clinically dense scenarios, while multi-type utterances are less frequent (5.3% vs. 8.0%). Per-type proportions broadly align: agenda_item is identical at 6.6%, detail is lower in synthetic data (36.4% vs. 46.9%), and question is higher (40.8% vs. 29.7%), consistent with the unanswered-question design objective. Critically, the synthetic corpus contributes 2.7% question_unanswered labels, entirely absent from the real annotations, providing the sole

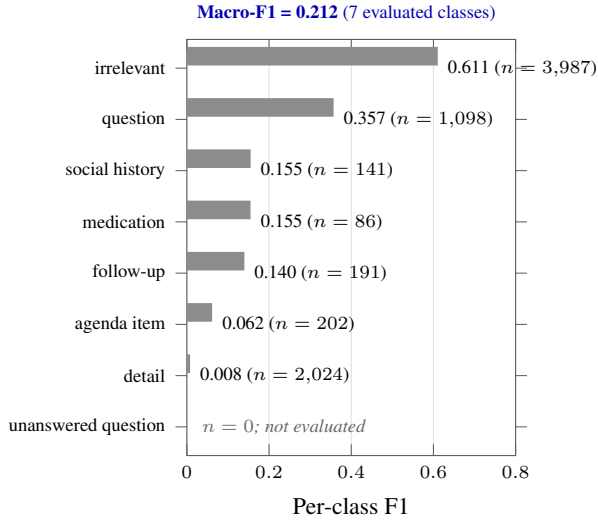


Figure 10: Zero-shot C1 performance by type on the pooled evaluation utterances ($n = 7,735$). Bars show per-class F1, with class support in parentheses. The displayed macro-F1 averages the seven evaluated classes (0.212). The evaluation set contains no question_unanswered examples; if this unsupported class is counted as F1=0, the eight-class mean is 0.186. Zero-shot prompting is particularly weak on agenda_item (0.062) and detail (0.008).

training signal for this category.

Naturalness gap Disfluency markers present the largest domain gap (Figure 12b): real conversations average 2.11 fillers per utterance versus 0.62 synthetically, and 29.4% of real patient turns are short responses (≤ 5 words) compared to 1.0%. Self-corrections remain comparable (0.045 vs. 0.039 per utterance). These gaps are expected from LLM-generated text and represent a known limitation; however, the primary role of this corpus is to supply question_unanswered exemplars rather than to replicate surface-level disfluency patterns.

Structural fidelity Mean turn counts are closely matched (D2N 55.9 vs. ST 54.8), though real conversations show higher variance ($\sigma=23.0$ vs. 8.7) and outliers exceeding 110 turns (Figure 12c). The main divergence is utterance length: real transcripts exhibit a heavy right-skew with physician turns reaching >200 words, while synthetic turns concentrate in the 5–25 word range (Figure 12d), yielding a lower length CV (1.82 vs. 0.30). Lexical diversity is comparable: unique bigram ratios nearly coincide (0.377 vs. 0.274) and global MATTR is modestly higher in synthetic data (0.666 vs. 0.734), reflecting less repetitive phrasing than spontaneous speech.

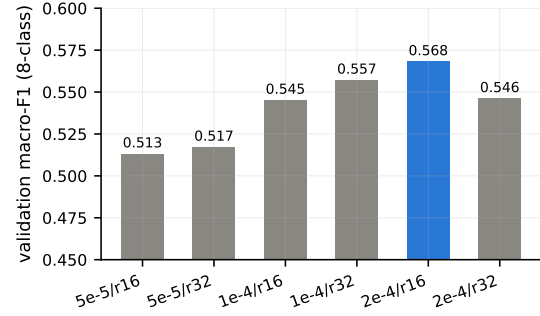


Figure 11: C1 hyperparameter sweep: learning rate $\times$ LoRA rank, four epochs with per-epoch checkpoints (seed 42), selected by validation macro-F1. The chosen configuration is $2 \times 10^{-4}/r16$.

H Formal Agenda-Status Semantics

For agenda a , let $D_t(a)$ be its associated details, $Q_t(a)$ its associated questions, and $R_t(a)$ its exact question-to-answer links at turn t . The agenda is *addressed* once any evidence of engagement exists ($|D_t(a)| > 0$ or $|R_t(a)| > 0$), and *fully addressed* when every question in $Q_t(a)$ has an answer link (or, when no questions exist, when at least one related detail exists). The derived per-agenda state is $z_t(a) \in \{\text{UNADDRESSED, ADDRESSED, FULLYADDRESSED}\}$, a deterministic function of recorded events and links.

I Evaluation Metrics

C1 is scored by per-class precision, recall, and F1 over the eight output classes (six event types, question_unanswered, irrelevant), macro-F1 over eight and over seven classes (excluding the near-empty question_unanswered), and conditional summary quality (ROUGE-1/2/L, BERTScore F1) against machine-label references. **The seven-class protocol is primary: question_unanswered is defined encounter-wide, so it is not a causal per-utterance prediction target, and its natural support is zero. The eight-class score is reported for completeness.** **C2** is scored by agenda-opening recall and delay, child parent-link accuracy (overall and given the annotated parent open), and resolution-state accuracy and balanced accuracy against manual labels. **C2q** is scored by balanced answerability accuracy, exact answer-utterance ID accuracy, answered/unanswered recall, and LLM verifier calls. **C3** is scored on summary-oriented, agenda-oriented, and efficiency axes. Statistical reporting (conversation-level bootstrap in-

tervals, paired policy comparisons, multiple seeds) accompanies the final tables.

Choice of reported C3 metrics. The C3 tables report four summarization metrics and six extraction metrics. Among the individual detection categories, we report Agenda Item, Detail, Question, and Irrelevant F1. Medication, Social-history, and Follow-up occur infrequently, making their individual F1 scores highly sensitive to a small number of predictions; their contributions are retained through Macro-F1, which averages performance across all detection categories.

J C3 Training and Evaluation Details

Training configuration. The C3 pairwise MLP is optimized with Adam using a learning rate of 1×10^{-3} , a weight decay of 1×10^{-4} , and a batch size of 256. The model minimizes mean squared error between its predictions and the attention-derived within-timestep percentile ranks. Training proceeds for up to 100 epochs with early stopping based on validation loss using a patience of 10 epochs. Training and validation examples are split by conversation with random seed 0, preventing source–target pairs from the same conversation from appearing in both splits.

Training data. The training set contains 23,409 source–target pairs. Each example pairs a buffered source summary with the target summary generated at the current timestep. Source and target summaries are encoded using normalized all-MiniLM-L6-v2 embeddings. The remaining auxiliary features encode source position, recency, speaker identity, token count, medical-entity count, and event type. The pairwise representation and attention-derived targets are defined in Appendix O.

Evaluation protocol. For extrinsic evaluation, C1 is held fixed while only the context-management strategy is varied. We compare the learned policy with FIFO, sliding-window, random, attention-based eviction, and an unlimited-buffer reference condition. FIFO evicts the oldest summaries when the token budget is exceeded, whereas the sliding-window strategy retains only the most recent fixed number of summaries before enforcing the token budget. Random eviction assigns random retention scores, and the attention-based strategy evicts summaries receiving the lowest C1 attention at the current timestep.

All strategies are evaluated independently under $B \in \{64, 100, 128, 256\}$ on the same 120 held-out conversations using the same frozen C1 checkpoint, atlas-c1-v3-llama-3.2-3b. Complete results across all strategies, budgets, and metrics are reported in Table 8.

K Extended C2 Results

In the predicted-C1 integrated run, 6.5% of 2,090 child events arrive before their annotated parent agenda has been opened and 17.1% are mislinked despite an open parent (Figure 9B). Separately, in reference-fed standalone C2, the tracker update raises non-null parent exact-ID accuracy from 0.837 to 0.896 over all 207 conversations and from 0.830 to 0.886 on the 140 non-training conversations. This gain comes at the cost of more over-linking when the annotated parent is null: null-abstention accuracy falls from 0.515 to 0.415. Null abstention counts the 1,732 children whose annotated parent is null (5,922 total minus the 4,190 with a non-null parent) and scores the fraction the tracker also leaves unlinked. The tracker leaves a child unlinked only when no agenda node has been opened yet, so its abstentions are confined to events preceding the first agenda opening. For reporting, nodes marked *resolved* internally count as *discussed*; the resulting two-state labels reach 0.903 balanced accuracy against manual labels.

L Extended C2q Routing Analyses

Held-out statistical validation (2026-08-03). Frozen-cache rows for 140 non-training conversations (20 validation + 120 test; 67 excluded; 139 contributing; patient/physician clusters: 58/136) were rebuilt without re-inference. Paired conversation-cluster bootstrap uses 10,000 resamples (seed 42, percentile 95% CIs; patient BA: 9,980 valid); exact two-sided McNemar uses six pairs per role/endpoint with within-family Holm correction. Exact-ID is answered-only ($n = 89/1,203$).

Confidence–verifier exact-ID is nonsignificant (adjusted $p = .832/.220$ for patient/physician). Confidence–NLI passes Holm for both endpoints/roles, but answerability McNemar tests question correctness, not balanced accuracy (BA): paired BA CIs include zero for both roles, whereas exact-ID CIs do not. Length–confidence BA CIs also include zero; McNemar instead favors confidence in correctness ($p = .0079/2.4 \times 10^{-15}$), not

Table 7: Held-out-140 bootstrap 95% CIs (patient: $n = 95/58$ clusters; physician: $n = 1,219/136$). Exact ID is answered-only.

Route	Balanced acc.	Exact ID (answered)
<i>Patient questions</i>		
NLI only	.737 [.544, .863]	.640 [.525, .758]
Verifier only	.755 [.532, .935]	.798 [.700, .892]
Length routing	.803 [.750, .858]	.607 [.500, .716]
Confidence routing	.727 [.508, .910]	.775 [.692, .860]
<i>Physician questions</i>		
NLI only	.655 [.528, .781]	.695 [.658, .730]
Verifier only	.680 [.552, .807]	.796 [.770, .821]
Length routing	.704 [.580, .820]	.728 [.698, .757]
Confidence routing	.683 [.557, .807]	.812 [.785, .839]

length’s BA.

Answer positions. Among the 183 answered patient questions, 174 (95.1%) are answered at the very next utterance; among the 2,261 physician questions, 2,232 (98.7%) are answered within four utterances of being asked. **These are annotation-layer statistics over the full 207-conversation corpus, not routing results.**

Candidate ordering. An archived four-way run with Top-2 candidates ranked by the HGB edge score leaves verifier-only routing essentially unchanged (≤ 0.001 on every metric) but shifts confidence routing. Distance order wins physician balanced accuracy 0.730 vs. 0.652; HGB reaches exact-ID 0.847 only by collapsing physician unanswered recall to 0.414, while distance order concedes little on the patient side (balanced 0.800 vs. 0.816).

The remaining analyses in this section (Top-1 versus Top-2, band selection, and band sensitivity) use the full 207-conversation corpus, which includes the verifier’s 67 training conversations, and are diagnostics only (§7.3).

Top-1 versus Top-2. Top-2 is not uniformly better. With the fixed band, moving from Top-1 to Top-2 raises exact ID from .738 to .798 for patient questions and from .819 to .825 for physician questions, but balanced accuracy changes from .803 to .800 and from .757 to .730, respectively, while verifier calls increase (32→43 and 378→454). Under fold-external validation selection, Top-2 again improves exact ID, but physician balanced accuracy falls from .731 to .687. Top-2 therefore improves exact-link coverage at the cost of more calls and, in some settings, unanswered performance.

Fixed versus validation-selected bands. For Top-2 confidence routing, validation selection

slightly improves patient balanced/exact accuracy (.800/.798→.808/.809) with 43→60 calls. The direction reverses for physician questions: the fixed band is better (.730/.825 versus .687/.776) and uses fewer calls (454 versus 813). Fold-external selection thus does not consistently improve [delete: held-out]out-of-fold performance.

Band sensitivity. Figure 8 shows that answerability, exact linking, and verifier cost trade off rather than improve together. For example, on physician Top-2, moving from the fixed band (.10, .75) to the post-hoc balanced-accuracy maximum (.20, .95) raises balanced accuracy from .730 to .778, but lowers exact ID from .825 to .754 and more than doubles verifier calls from 454 to 963. We therefore retain the fixed band for reporting rather than promote a full-plate maximum.

M Extended C3 Commentary

The divergence between summary similarity and agenda detection suggests differing evaluation sensitivities. Summary-similarity metrics reward the recovery of lexical and semantic content that appears in the reference summary, including information that may recur across prior summaries. In contrast, agenda detection requires the model to preserve distinctions among event types and identify whether a specific agenda item is present. Additional context may therefore increase the amount of reference-related information available to the model while also introducing redundant or competing evidence that weakens event-level discrimination. This interpretation is consistent with the increase in all four summary-similarity metrics and the aggregate decline in agenda-oriented performance as the budget grows. The present evaluation does not isolate redundancy or interference experimentally, so these mechanisms should be treated as explanations consistent with the observed pattern rather than as directly established causal effects.

Sliding and FIFO expose a trade-off between coverage and context concentration. FIFO uses the available token budget to preserve as much accumulated context as possible, including older summaries that may still contribute lexical or semantically related content to the generated summary. Sliding instead imposes a recency constraint and discards older summaries even when additional tokens remain available. At $B = 256$, Sliding obtains numerically higher scores on Agenda Com-

Table 8: Full extrinsic evaluation results across context-selection policies and token budgets. Bold values indicate the best result among the five budget-constrained policies at each budget. The no-eviction condition retains the full buffer and is therefore not included when determining the best budget-constrained result. AC denotes Agenda Completeness, AI denotes Agenda Item F1, and Macro denotes Macro-Average F1.

<i>B</i>	Policy	R-1	R-2	R-L	BERT	AC	AI F1	Detail F1	Question F1	Irrelevant F1	Macro F1
64	Random	0.6849	0.4896	0.6589	0.8700	0.4922	0.4980	0.7272	0.7835	0.7547	0.4842
	FIFO	0.6862	0.4875	0.6602	0.8711	0.5016	0.5148	0.7475	0.7769	0.7678	0.4903
	Sliding	0.6862	0.4875	0.6602	0.8711	0.5016	0.5148	0.7475	0.7769	0.7678	0.4903
	Attention	0.6861	0.4904	0.6602	0.8706	0.4878	0.5131	0.7438	0.7779	0.7648	0.4881
	Learned	0.6870	0.4931	0.6620	0.8714	0.5186	0.4886	0.7488	0.7761	0.7613	0.4871
	No eviction	0.6928	0.4994	0.6676	0.8734	0.4787	0.4931	0.7192	0.7611	0.7701	0.4855
100	Random	0.6893	0.4947	0.6651	0.8725	0.5067	0.4824	0.7245	0.7701	0.7669	0.4851
	FIFO	0.6920	0.4964	0.6675	0.8735	0.5045	0.4914	0.7383	0.7739	0.7733	0.4860
	Sliding	0.6921	0.4963	0.6676	0.8735	0.5045	0.4914	0.7381	0.7727	0.7729	0.4854
	Attention	0.6890	0.4929	0.6637	0.8719	0.5081	0.4969	0.7374	0.7778	0.7740	0.4864
	Learned	0.6901	0.4954	0.6647	0.8721	0.5033	0.5197	0.7419	0.7739	0.7722	0.4899
	No eviction	0.6928	0.4994	0.6676	0.8734	0.4787	0.4931	0.7192	0.7611	0.7701	0.4855
128	Random	0.6883	0.4935	0.6637	0.8715	0.4987	0.4943	0.7215	0.7712	0.7654	0.4885
	FIFO	0.6923	0.4979	0.6685	0.8736	0.4965	0.5031	0.7302	0.7732	0.7724	0.4863
	Sliding	0.6893	0.4946	0.6651	0.8726	0.5081	0.4987	0.7325	0.7723	0.7719	0.4869
	Attention	0.6923	0.4980	0.6679	0.8731	0.4951	0.5161	0.7230	0.7703	0.7670	0.4898
	Learned	0.6898	0.4941	0.6650	0.8724	0.4983	0.4969	0.7331	0.7659	0.7738	0.4861
	No eviction	0.6928	0.4994	0.6676	0.8734	0.4787	0.4931	0.7192	0.7611	0.7701	0.4855
256	Random	0.6933	0.5002	0.6683	0.8737	0.4859	0.4886	0.7197	0.7601	0.7694	0.4839
	FIFO	0.6937	0.5000	0.6690	0.8742	0.4842	0.4841	0.7203	0.7618	0.7708	0.4852
	Sliding	0.6919	0.4966	0.6677	0.8732	0.5059	0.4923	0.7259	0.7688	0.7737	0.4853
	Attention	0.6944	0.5006	0.6693	0.8741	0.4816	0.4833	0.7175	0.7640	0.7682	0.4846
	Learned	0.6947	0.5016	0.6700	0.8744	0.4809	0.4837	0.7194	0.7634	0.7699	0.4826
	No eviction	0.6928	0.4994	0.6676	0.8734	0.4787	0.4931	0.7192	0.7611	0.7701	0.4855

pleteness and all five detection-oriented F1 metrics while using fewer buffer tokens. This pattern is consistent with the possibility that concentrating the context on a smaller set of recent summaries reduces competition among agenda signals. Conversely, FIFO’s small advantage on all four summary-similarity metrics suggests that some older information removed by Sliding remains useful for reproducing the reference summary. The comparison therefore indicates that context efficiency depends not only on token count but also on whether the retained evidence is concentrated around the current conversational state.

No eviction most consistently harms agenda completeness. Although the no-eviction condition retains every accumulated summary, its disadvantage is not uniform across all category-level F1 metrics. Its most consistent disadvantage appears in Agenda Completeness, on which it ranks below every budget-constrained condition, whereas its rankings on Agenda Item F1 and Macro F1 are more mixed. This pattern indicates that unlimited context does not degrade every prediction

uniformly. A long buffer may instead make some previously introduced agenda items less salient relative to duplicated, recent, or semantically overlapping summaries. The no-eviction condition is therefore better interpreted as evidence that complete retention does not ensure complete agenda recovery than as a universally inferior condition on every metric.

N Verifier Route Analysis: Full Accounting

Across the 43 questions labeled unanswered (14 patient / 29 physician), human review identifies 18 off-target replies, 14 counter-questions, 3 backchannels, 1 deferral, and 7 other non-answers; all 43 rows are labeled. The validation-selected NLI-only policy abstains correctly on 34 and the confidence-routing cascade on 27: the transition rescues 2 patient rows and regresses 9 physician rows. The verifier itself rejects what it is shown (12 rows terminate as direct verifier-NO abstentions), but the confidence band routes around it: 9 rows are accepted directly on high NLI similarity without ver-

ification, and 7 receive an incorrect verifier YES. Regenerating the analysis with the human-reviewed labels (2026-07-28) leaves every aggregate count unchanged and yields the per-type accounting: the 2 rescued rows are patient counter-questions; the 9 regressions are physician-side (4 off-target, 1 counter-question, 4 other); the 9 direct NLI acceptances split 4 off-target / 3 counter-question / 2 other; and the 7 incorrect verifier accepts split 4 off-target / 1 counter-question / 2 other. These are policy-level comparisons, not a verifier-only causal effect, since the compared policies also differ in thresholds.

Beyond the frozen window. A full-transcript review extends the answer search past lookahead 4 for the 43 labeled-unanswered questions (Figure 13). Eleven of the 14 patient questions and 8 of the 29 physician questions receive an answer later in the encounter, at distances up to 90 utterances. The lookahead-4 labels are therefore window-relative: they mark questions whose answers a bounded streaming linker cannot reach in time, and 3 patient and 21 physician questions receive no answer anywhere in the transcript.

Statistical notes. The unanswered denominators are small, so we report exact uncertainty for the key rates (Wilson 95% intervals): patient unanswered recall $11/14 = 0.786$ [0.524, 0.924]; physician unanswered recall $17/29 = 0.586$ [0.407, 0.745]; NLI-only correct abstentions $34/43 = 0.791$ [0.648, 0.886]; cascade correct abstentions $27/43 = 0.628$ [0.479, 0.756]. Treating the policy transition as paired decisions over the same 43 questions (2 rescued vs. 9 regressed discordant rows), an exact McNemar test gives two-sided $p = 0.065$: the abstention difference between NLI-only and confidence routing is not statistically significant at $\alpha = 0.05$ on this denominator, consistent with reporting unanswered recall as preliminary.

O Extended Analysis of BERTScore Hindsight versus Attention Supervision

Table 9 compares two complete C3 policy designs. BERT-396 is the original source-only policy trained with binary hindsight labels based on the maximum future BERTScore, whereas Attn-780 is the proposed source–target pairwise policy trained with within-timestep attention percentile

ranks. Both policies are evaluated on the same 120-conversation test set using the same frozen C1 model (atlas-c1-v3-llama-3.2-3b); only the context-selection policy differs.

Policy formulations. BERT-396 represents each buffered source summary s_i using a 396-dimensional feature vector:

$$\mathbf{x}_{i,t}^{\text{BERT}} = [\mathbf{e}(s_i); \mathbf{m}_{i,t}] \in \mathbb{R}^{396}, \quad (4)$$

where $\mathbf{e}(s_i) \in \mathbb{R}^{384}$ is the source-summary embedding and $\mathbf{m}_{i,t} \in \mathbb{R}^{12}$ contains the auxiliary features. These features encode the source position, recency, speaker, token count, medical-entity count, and event type.

The policy is trained using a binary hindsight target:

$$y_{i,t} = \mathbb{I} \left[\max_{j \geq t} \text{BERTScore}(s_i, s_j) > 0.75 \right], \quad (5)$$

which labels a buffered summary as relevant when it is sufficiently similar to at least one future summary.

Attn-780 additionally conditions the retention score on the target summary s_t produced at the current timestep:

$$\mathbf{x}_{i,t}^{\text{ATTN}} = [\mathbf{e}(s_i); \mathbf{e}(s_t); \mathbf{m}_{i,t}] \in \mathbb{R}^{780}. \quad (6)$$

The pairwise MLP predicts the source summary’s attention-derived within-timestep percentile rank:

$$\hat{r}_{i,t} = f_{\theta}(\mathbf{x}_{i,t}^{\text{ATTN}}). \quad (7)$$

Percentile ranks are used because the eviction decision depends on the relative ordering of the summaries available at the same timestep. Raw attention shares are less suitable because they sum to one within each buffer and therefore vary systematically with buffer size.

Results at representative budgets. At $B = 128$, Attn-780 outperforms BERT-396 on all ten primary metrics. The largest absolute improvements occur in Irrelevant F1 (0.7560 $\rightarrow$ 0.7738), Detail F1 (0.7159 $\rightarrow$ 0.7331), and Agenda Item F1 (0.4805 $\rightarrow$ 0.4969). Macro-F1 also increases from 0.4794 to 0.4861. The four summary-similarity metrics improve by smaller margins, but all move in the same direction.

At $B = 256$, Attn-780 remains stronger on eight of the ten metrics. It outperforms BERT-396 on all four summary-similarity metrics and on Detail F1,

Table 9: Comparison of hindsight- and attention-supervised C3 policies on the 120-conversation test set. Bold indicates the better result at the same budget. BERT-396 uses binary BERTScore-0.75 hindsight labels, whereas Attn-780 uses within-timestep attention percentile ranks.

Policy	Budget	R-1	R-2	R-L	BERT	AC	AI-F1	D-F1	Q-F1	I-F1	Macro-F1
BERT-396	128	0.6890	0.4938	0.6638	0.8720	0.4936	0.4805	0.7159	0.7584	0.7560	0.4794
Attn-780	128	0.6898	0.4941	0.6650	0.8724	0.4983	0.4969	0.7331	0.7659	0.7738	0.4861
BERT-396	256	0.6940	0.5009	0.6695	0.8738	0.5023	0.4843	0.7152	0.7632	0.7669	0.4815
Attn-780	256	0.6947	0.5016	0.6700	0.8744	0.4809	0.4837	0.7194	0.7634	0.7699	0.4826

AC: Agenda Completeness; AI-F1: Agenda Item F1; D-F1: Detail F1; Q-F1: Question F1; I-F1: Irrelevant F1.

Question F1, Irrelevant F1, and Macro-F1. BERT-396 performs better on Agenda Completeness, with 0.5023 compared with 0.4809, and is marginally better on Agenda Item F1, with 0.4843 compared with 0.4837.

Interpretation. The advantage of Attn-780 is more pronounced when the context budget is restrictive and narrows as more context can be retained. This result is consistent with the hypothesis that similarity to future summaries is an incomplete proxy for identifying the context actually utilized by C1. A buffered summary may be semantically similar to later content without being useful for the current state update, particularly when clinical details are repeated across multiple turns.

Attention-derived supervision instead provides a timestep-specific signal of how strongly C1 uses each buffered summary while generating the current output. Conditioning on both the source and target summaries can therefore support more discriminative retention decisions when the policy must choose among several competing context items under a tight budget.

The results at $B = 256$ also show that this advantage is not uniform across all task-oriented metrics. In particular, Attn-780’s lower Agenda Completeness suggests that attention utilization does not necessarily capture every item that should remain available for long-term agenda coverage. Attention-derived selection is therefore most beneficial under tight context constraints, whereas hindsight similarity can retain a small advantage for preserving broader agenda coverage when the available budget is larger.

Scope of the comparison. This experiment changes both the policy representation and its supervision. BERT-396 uses a source-only representation with binary hindsight labels, whereas Attn-780 uses a source–target representation with continuous attention-rank targets. The experiment

therefore compares two complete C3 policy designs rather than isolating the causal contribution of attention supervision alone. A controlled label-only or representation-only ablation would be required to separate these effects.

P Patient–Provider Conversation Dynamics: Extended Analysis

The encounter is physician-driven (§3): physician questions operate as history-taking machinery (asked and answered within a turn or two almost without exception), while patient questions are rare, resolve more slowly, and have a higher observed unanswered rate in the frozen lookahead-4 annotations (7.1% versus 1.3%).

The failure modes are asymmetric in an informative way. When a physician’s question fails, the patient has usually *tried* to answer: the reply is topically adjacent but off-target, cooperative but imprecise (15 of 29 reviewed physician non-answers are off-target, versus 3 of 14 on the patient side; the physician asks for a precise clinical attribute and receives approximate lay information, as when a request for daily glucose readings is answered with “pretty good”). When a patient’s question fails, the physician has usually *not* tried: the question is met with a counter-question, a backchannel, or a deferral that the visit never returns to. A plausible reading is floor control: patients answer when asked, even imperfectly, while physicians manage the visit’s agenda and defer, and deferred items then compete with everything else in a time-limited encounter.

These dynamics carry design consequences. High-similarity non-answers concentrate on the physician side, and they remain the binding failure mode: the route analysis (Appendix N) shows the confidence band accepts a share of them outright on similarity alone and the verifier wrongly accepts others, so confidence routing trades physician unanswered recall for answered-side accuracy rather

Table 10: Results after removing the reference answer (lookahead = 4). Values are n/N (%). In the physician removal row, the second percentage excludes cases in which the model selected a physician utterance.

Measure	GPT-5.5	Claude Opus 4.8
Reference answer selected (answered rows)	2,180/2,232 (97.7%)	2,174/2,232 (97.4%)
Unanswered recall (physician)	18/29 (62.1%)	19/29 (65.5%)
Selected after removal (physician)	429/2,232 (19.2%; 14.3%)	360/2,232 (16.1%; 12.5%)
Selected after removal (patient)	8/183 (4.4%)	10/183 (5.5%)

Table 11: C1 macro-F1 across training generations on the fixed evaluation pool ($n = 7,735$). The 7-class score excludes the near-empty question_unanswered class.

Model	7-class	8-class
Zero-shot	0.212	0.186
V2 (QLoRA)	0.676	0.592
V3 (V2 + synthetic)	0.717	0.628

than eliminating the failure. Deflections are usually low-similarity and rejected upstream; the cascade’s clearest abstention gains (its only rescued rows) are patient counter-questions, so on the patient side calibrated abstention is both the binding constraint and where the design helps most. They also locate the system’s value: physician questions largely resolve themselves, so an agenda tracker changes outcomes chiefly by protecting the patient’s side of the conversation. We note the caveat that these are observational regularities of one ACI-BENCH-derived corpus of primary-care-style encounters; the asymmetries may shift across specialties, settings, and visit types, and nothing here supports a causal reading.

Q Frontier-Model Answer-Removal Sensitivity Analysis

In the physician-question ablation (Table 1), verifier-only (Agenda_LLM-only) routing raises exact-ID accuracy and answered recall relative to NLI-only routing, but lowers unanswered recall and balanced accuracy. For the additional comparison in Table 10, we count whether each model selects a frozen reference answer when one is present, reuse unanswered recall for the 29 naturally unanswered physician questions. We then remove every frozen reference answer utterance from paired answered windows and count how often the model still selects another utterance. Both models usually select the frozen reference answer when it is present, but physician unanswered recall remains 62–66%. After removal, they still select another utterance in 16–19% of physician windows and 4–5% of patient windows. The altered windows were not independently re-annotated, so this is a sensitivity analysis rather than additional unanswerable annotations. The pattern agrees with prior abstention and evidence-replacement evaluations (Madhusud-

han et al., 2025; Lavi et al., 2026) and supports confidence routing that limits LLM verification to uncertain cases.

R Extended Results and Reproducibility

C1 recipe deltas. Relative to V2, the V3 recipe raises 7-class macro-F1 from 0.676 to 0.717 on the fixed 7,735-utterance evaluation pool. The gain concentrates in medication (+0.225) and agenda item (+0.087) while the remaining types are flat or slightly down: a subset-driven shift rather than a uniform one (Figure 6).

The full 684-row threshold-band sweep behind Figure 8, the per-class zero-shot scores plotted in Figure 10, and the six-policy $\times$ four-budget context grids summarized in §7.4 are released as CSV artifacts with manifests recording code revision, model revisions, annotation hashes, and fold assignments. All C2q evaluations are five-fold encounter-family cross-fitted (seed 42, zero family and zero conversation overlap) with fixed thresholds that are never tuned on scored rows.

Table 3 caveats. Columns: Status is status macro-F1, Link is parent-linking accuracy, First is first-mention accuracy, and Pat./Phys. are balanced answerability for patient and physician questions. Error attribution under reference C1: 59 of 75 wrong statuses (79%) trace to a wrong or missing answer link, 15 to a wrong parent, and none to missed openings. The balanced-answerability columns use the plate-restricted denominators of 95 patient and 1,219 physician questions, and the status macro-F1 is computed on a strict 144-node subset described below. The First column is raw first-mention accuracy over an imbalanced label set (4,286 true / 87 false); the corresponding balanced accuracies are 0.580, 0.338, and 0.375 for the three rows. The bounded budget is nominal: a single summary longer than 64 tokens is retained rather than truncated (16 reference / 20 predicted turns exceed it, at most 87 / 110 tokens). The held-out plate contains 280 agenda nodes: every agenda_item

annotation in the 140 held-out conversations, with multiple agenda items on one utterance counted separately. Sixteen nodes are excluded because their reference status cannot be determined unambiguously (multi-node parent ambiguity or questions labeled Exclude), leaving a strict pool of 264; 144 of these (54.5%, skewed 128/15/1 toward fully-addressed) have a determinate status under all four combinations of C1 input and context, and the other 120 are indeterminate in at least one condition through an ambiguous predicted agenda identifier or a generated or out-of-plate question with no corresponding C2q prediction. Promoting the status F1 to a primary result requires the child-to-agenda sidecar (or an approved exclusion policy), C2q rows for predicted false-positive questions, and row-level frozen-protocol predictions; no GPU re-run is needed.

S Implementation Details

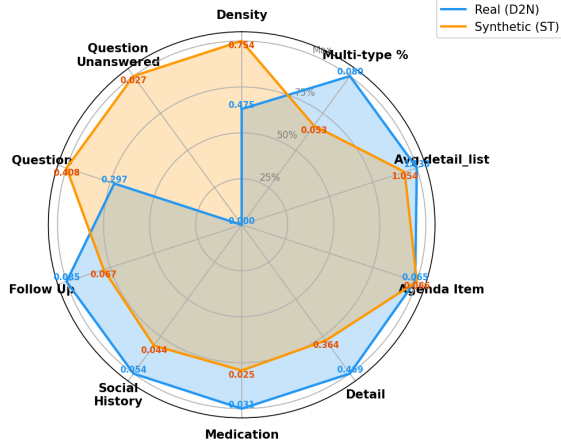
C1 fine-tunes Llama-3.2-3B with 4-bit QLoRA (rank 16, learning rate 2×10^{-4} , seed 42), selected by the sweep of Figure 11: three learning rates $\{5 \times 10^{-5}, 1 \times 10^{-4}, 2 \times 10^{-4}\}$ crossed with two adapter ranks $\{16, 32\}$, each trained for four epochs with per-epoch checkpoints. Training data combines the 67 real training conversations (3,743 rows) with 100 synthetic conversations (5,478 rows) into a 9,221-row base, expanded to 16,436 examples by three augmentations: eightfold replication of the 369 multi-label utterances (+2,583), context-erased copies (+2,056), and supplementation of irrelevant rows to a fixed 64.2% ratio (+2,576). The C2q NLI scorer is the cross-encoder/nli-MiniLM2-L6-H768 cross-encoder, and the yes/no verifier reuses the frozen C1-v3 weights. C3 embeds summaries with the frozen all-MiniLM-L6-v2 encoder (L_2 -normalized) and tags entities with scispaCy (Neumann et al., 2019) (en_core_sci_sm, v0.5.4). Within its 780-dimensional pairwise feature vector, position and recency are defined as $p_i = i/256$ and $r_{i,t} = (t - i)/t$, where i is the source-summary timestep and t the current timestep; the fixed denominator for position prevents it from being algebraically redundant with recency. The scorer is a 780–128–64–1 MLP with ReLU activations, trained with mean squared error to predict the within-timestep attention percentile rank. Training uses Adam with learning rate 1×10^{-3} , weight decay 1×10^{-4} , and batch size 256, for

up to 100 epochs with early stopping, on 23,409 source–target pairs split into training and validation at the conversation level to prevent leakage. Component evaluations ran on single consumer GPUs (RTX 4090 / A40); the end-to-end error analysis and streaming-trace run used an NVIDIA RTX PRO 6000 Blackwell server (97.9 GB; driver 580.159.03) with Python 3.12.3, PyTorch 2.8.0+cu128, and Transformers 5.1.0. Prompt templates for annotation, synthetic generation, and verification are included in the dataset release.

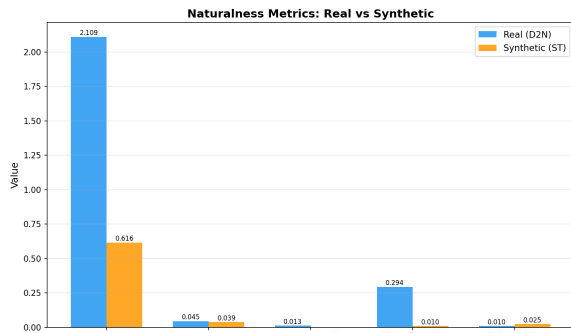
Disclosure of AI Use

Large language model tools assisted with editing and manuscript preparation. AI systems used in the pipeline and evaluation are documented where they appear. The authors remain fully responsible for all content.

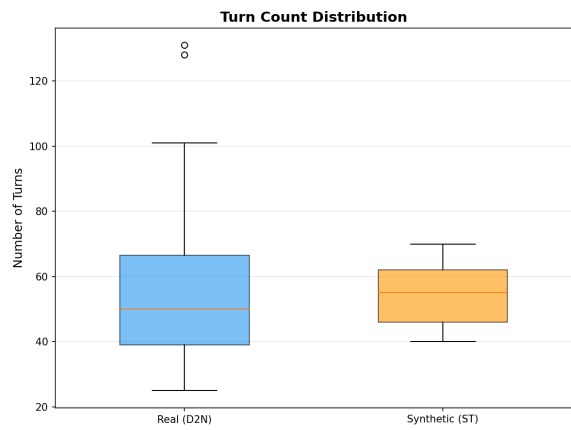
Radar: Annotation Distribution — Real vs Synthetic



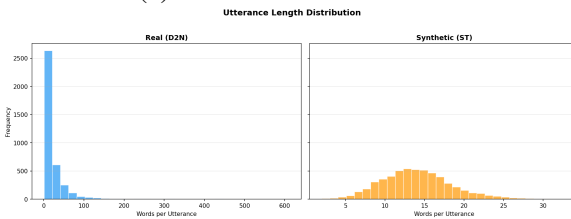
(a) Annotation distribution radar



(b) Naturalness metrics



(c) Turn count distribution



(d) Utterance length distribution

Figure 12: Synthetic (ST) vs. real (D2N) conversation comparison. (a) Annotation radar; question_unanswered exists only in synthetic data. (b) Naturalness gap, primarily in filler frequency and short-response rate. (c–d) Structural similarity in turn count but divergence in utterance length.

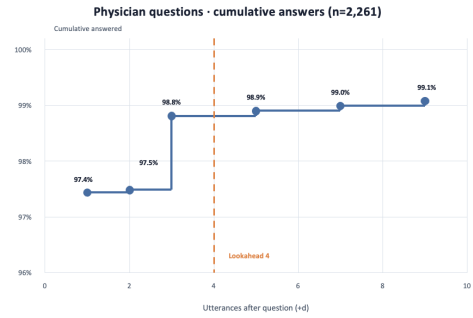
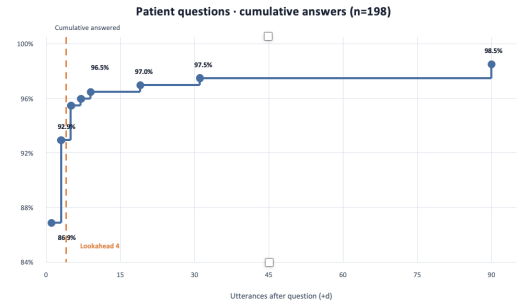


Figure 13: Cumulative share of questions answered by dialogue distance after the question, with the lookahead-4 boundary marked (top: patient, bottom: physician). A full-transcript review of the 43 questions labeled unanswered at lookahead 4 finds that 11 of 14 patient and 8 of 29 physician questions receive an answer later in the encounter, leaving 3 patient and 21 physician questions unanswered anywhere in the transcript.